

# Maximize Equipment Life

## Proactively Prevent Unplanned Downtime Caused by Component Failures and Major Repairs.

Hapman equipment is robustly engineered to run trouble-free for years. But it must be properly maintained to prevent component failures and major repairs from causing unplanned downtime. To keep maintenance on track, enroll in our Managed Maintenance Program (MMP). Each visit will start with a top-to-bottom inspection of your Hapman equipment by one of our experienced technicians. It will also include:

- Maintenance baseline for your Hapman equipment to discover any early warnings of part wear and/or impending component failures.
- Detailed inspection report pinpointing maintenance/repair needs, so that -if you choose- so they can be addressed before causing a problem.
- OEM maintenance expertise and recommendations concerning equipment maintenance, improvements, and modifications.

### Maintenance Program Options:

Hapman offers three Managed Maintenance Program (MMP) options based on the level of assistance you want. All three will improve your machine uptime, reliability, performance, and longevity.

**SEMI-ANNUAL:** Two (2) service visits per year by a Hapman technician at discounted MMP Rates. Discount on parts (10%) and standard services rates (15%) for all non-MMP visits.

**QUARTERLY:** Four (4) service visits per year by a Hapman technician at discounted MMP Rates. Discount on parts (15%) and standard services rates (20%) for all non-MMP visits.

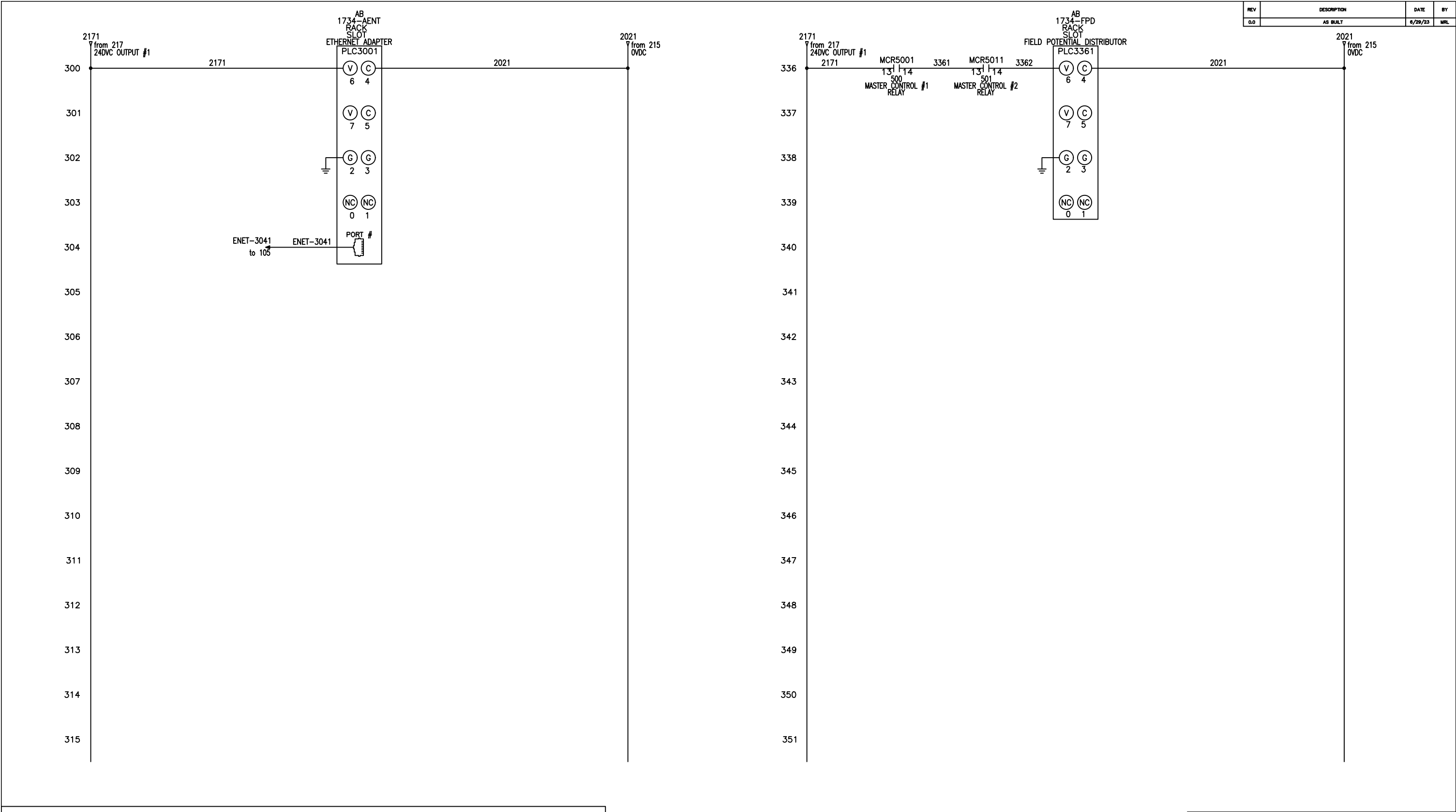
**MONTHLY/WEEKLY:** Multiple service visits per year by a Hapman technician. Increased custom discounts for parts and services.

**If you need technical support, give us a call. We are available 24/7.**

**Exclusive  
Savings!**  
When you sign up for  
the Hapman MMP!







2171

from 217

24VDC OUTPUT #1

2171

3361

3362

from 215

0VDC

336

337

338

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1734-FPD

RACK

SLOT

FIELD POTENTIAL DISTRIBUTOR

PLC3361

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V

C

7

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3

NC

NC

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MCR5001

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500

MASTER CONTROL #1

RELAY

MCR5011

13

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MASTER CONTROL #2

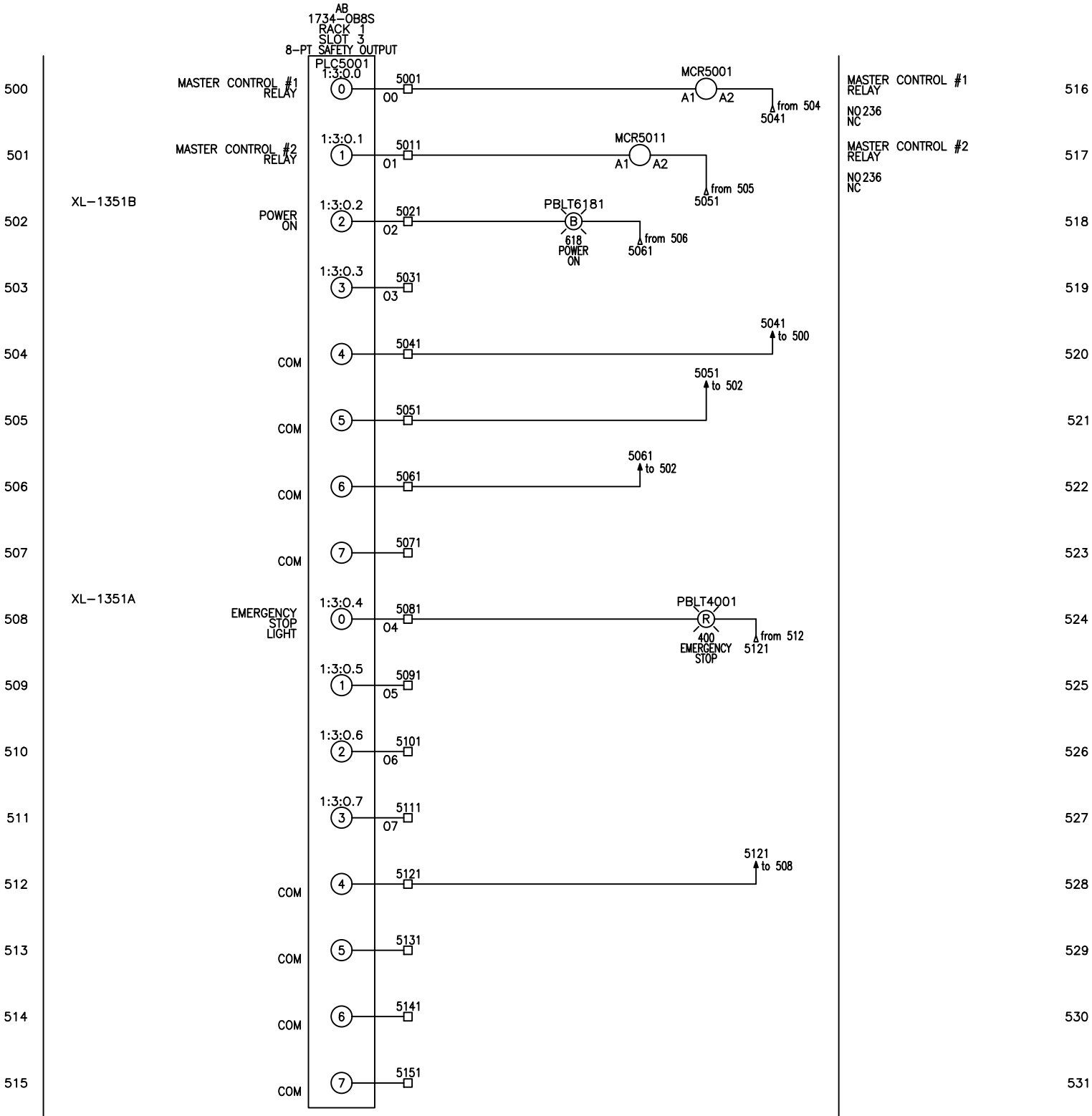
RELAY

| CONDUCTOR WIRE CODE |                                | WIRE GAUGE        |                | LEGEND |              |
|---------------------|--------------------------------|-------------------|----------------|--------|--------------|
| BLACK:              | GREATER THAN 120VAC            | MOTOR CIRCUITS:   | MIN. 12 AWG    | -----  | FIELD WIRING |
| RED:                | 120VAC OR LESS                 | CONTROL CIRCUITS: | MIN. 16 AWG    | _____  | PANEL WIRING |
| WHITE:              | GROUNDING AC LINE              | PLC I/O:          | MIN. 18 AWG    |        |              |
| BLUE:               | UN GROUNDED VDC                |                   |                |        |              |
| WH/BLU:             | GROUNDING VDC                  | FUSES:            | LINE/LOAD SIDE |        |              |
| GREEN:              | GROUNDING CONDUCTORS           |                   | SHALL BE EQUAL |        |              |
| ORANGE:             | CIRCUITS THAT REMAIN ENERGIZED |                   | AWG            |        |              |
|                     | WHEN MAIN DISC. IS "OFF"       | RETURN/SUPPLY:    | SHALL BE EQUAL |        |              |
|                     |                                |                   | AWG            |        |              |

|                                                                      |                          |                                                                                      |                |
|----------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------|----------------|
| <b>HAPMAN</b><br><small>ideas that move</small>                      |                          | 5944 E. N AVE., KALAMAZOO, MICHIGAN 49048<br>PHONE (269) 343-1675 FAX (269) 349-2477 |                |
| DATE<br>4/21/23                                                      | JOB NO.<br>X20335-AC     | DRAWN<br>BOM = 1                                                                     | MRL<br>DWG = 1 |
| TITLE<br><b>PLC WIRING DIAGRAM</b><br>TYPE 4X - HELIX CONVEYOR HI/LO |                          |                                                                                      |                |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                           |                          |                                                                                      |                |
| SHEET<br>3 OF 16                                                     | DWG. NO.<br>X20335AC-B00 | REV.<br>0.0                                                                          |                |



| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



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LEGEND

----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING

HAPMAN

ideas that move

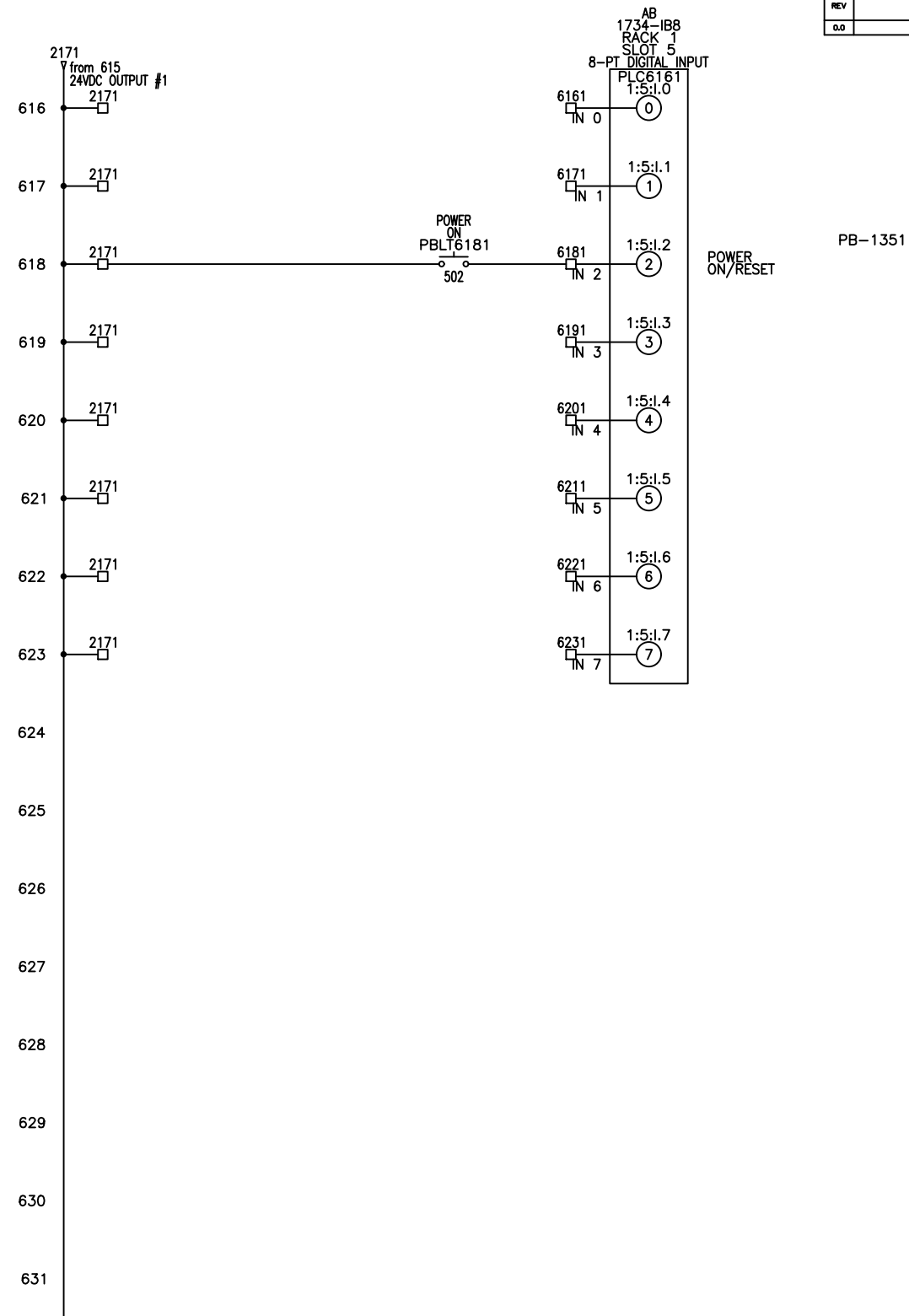
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PHONE (269) 343-1675 FAX (269) 349-2477

|              |                   |         |         |
|--------------|-------------------|---------|---------|
| DATE 4/21/23 | JOB NO. X20335-AC | DRAWN   | MRL     |
| SCALE N/A    |                   | BOM = 1 | DWG = 1 |

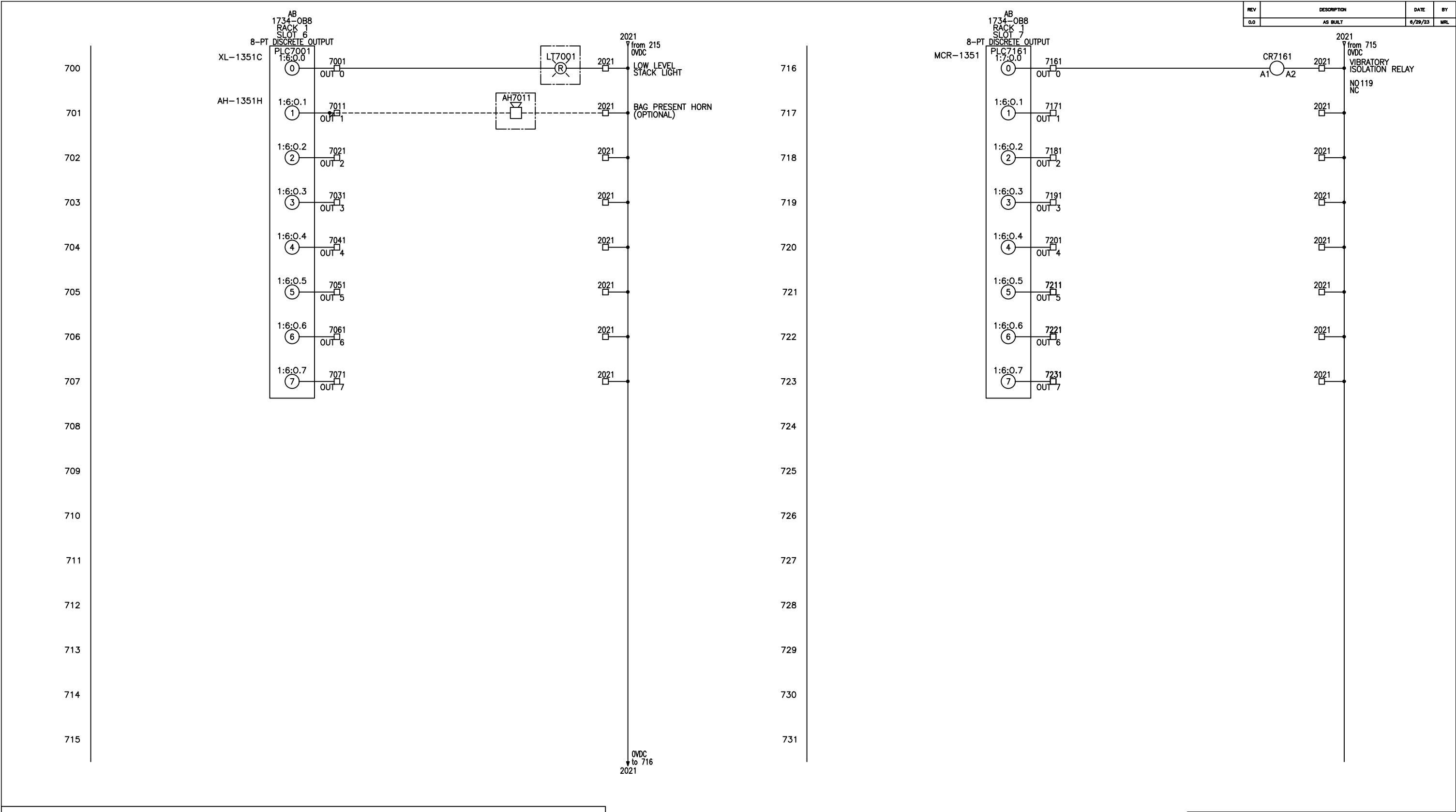
TITLE  
**PLC WIRING DIAGRAM**  
TYPE 4X - HELIX CONVEYOR HI/LO

|                                            |                       |
|--------------------------------------------|-----------------------|
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) | REV. 0.0              |
| SHEET 5 OF 16                              | DWG. NO. X20335AC-B00 |





|                                                                                                                                      |              |                                                                                      |         |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------------------------------------|---------|
|                                                 |              | 5944 E. N AVE., KALAMAZOO, MICHIGAN 49045<br>PHONE (269) 343-1675 FAX (269) 348-2477 |         |
| IDEAS THAT MOVE                                                                                                                      |              |                                                                                      |         |
| DATE                                                                                                                                 | 4/21/23      | JOB NO.                                                                              |         |
| SCALE                                                                                                                                | N/A          | X20335-AC                                                                            |         |
|                                                                                                                                      |              | DRAWN                                                                                | MRL     |
|                                                                                                                                      |              | BOM = 1                                                                              | DWG = 1 |
| TITLE<br><h2 style="text-align: center;">PLC WIRING DIAGRAM</h2> <h3 style="text-align: center;">TYPE 4X - HELIX CONVEYOR HI/LO</h3> |              |                                                                                      |         |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                                                                                           |              |                                                                                      |         |
| SHEET                                                                                                                                | 6            | OF                                                                                   | 16      |
| DWG. NO.                                                                                                                             | X20335AC-B00 |                                                                                      |         |
| REV.                                                                                                                                 | 0            |                                                                                      |         |



| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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ORANGE: CIRCUITS THAT REMAIN ENERGIZED WHEN MAIN DISC. IS "OFF"

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CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL AWG

LEGEND

----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING

**HAPMAN**  
ideas that move

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|                 |                      |         |         |
|-----------------|----------------------|---------|---------|
| DATE<br>4/21/23 | JOB NO.<br>X20335-AC | DRAWN   | MRL     |
| SCALE<br>N/A    |                      | BOM = 1 | DWG = 1 |

TITLE  
**PLC WIRING DIAGRAM**  
TYPE 4X - HELIX CONVEYOR HI/LO

|                                            |                       |
|--------------------------------------------|-----------------------|
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) | REV. 0.0              |
| SHEET 7 OF 16                              | DWG. NO. X20335AC-B00 |



| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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\_\_\_\_\_ PANEL WIRING



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|                                            |         |          |              |         |         |          |
|--------------------------------------------|---------|----------|--------------|---------|---------|----------|
| DATE                                       | 4/21/23 | JOB NO.  | X20335-AC    |         | DRAWN   | MRL      |
| SCALE                                      | N/A     |          |              | BOM = 1 | DWG = 1 |          |
| TITLE                                      |         |          |              |         |         |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |          |              |         |         |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |          |              |         |         |          |
| SHEET                                      | 8 OF 16 | DWG. NO. | X20335AC-B00 |         |         | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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\_\_\_\_\_ PANEL WIRING



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|                                            |         |         |           |          |              |          |
|--------------------------------------------|---------|---------|-----------|----------|--------------|----------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-AC |          | DRAWN        | MRL      |
| SCALE                                      | N/A     |         | BOM = 1   | DWG = 1  |              |          |
| TITLE                                      |         |         |           |          |              |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |          |
| SHEET                                      | 9       | OF      | 16        | DWG. NO. | X20335AC-B00 | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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\_\_\_\_\_ PANEL WIRING



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|                                            |         |         |           |          |              |          |
|--------------------------------------------|---------|---------|-----------|----------|--------------|----------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-AC |          | DRAWN        | MRL      |
| SCALE                                      | N/A     |         |           |          | BOM = 1      | DWG = 1  |
| TITLE                                      |         |         |           |          |              |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |          |
| SHEET                                      | 10      | OF      | 16        | DWG. NO. | X20335AC-B00 | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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SHALL BE EQUAL  
AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL  
AWG

LEGEND

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\_\_\_\_\_ PANEL WIRING



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|--------------------------------------------|---------|---------|-----------|----------|--------------|----------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-AC |          | DRAWN        | MRL      |
| SCALE                                      | N/A     |         | BOM = 1   | DWG = 1  |              |          |
| TITLE                                      |         |         |           |          |              |          |
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| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |          |
| SHEET                                      | 11      | OF      | 16        | DWG. NO. | X20335AC-B00 | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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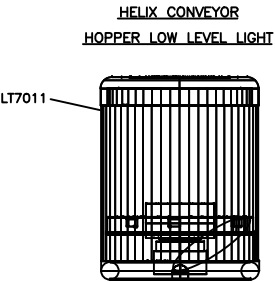
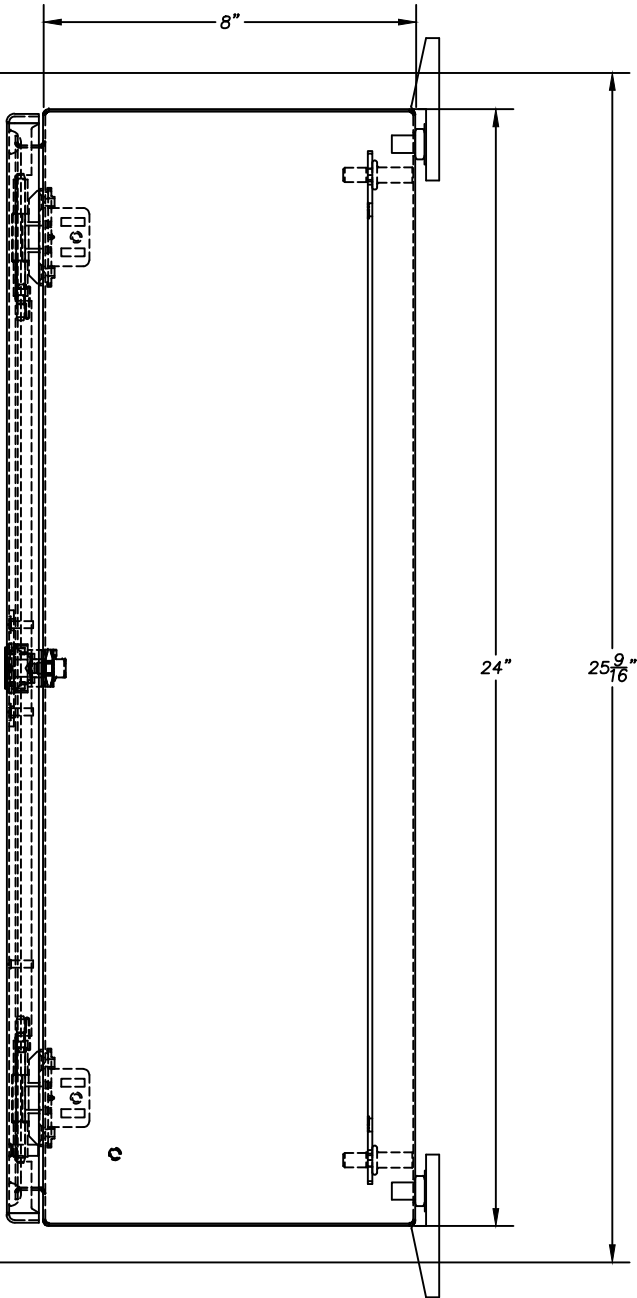
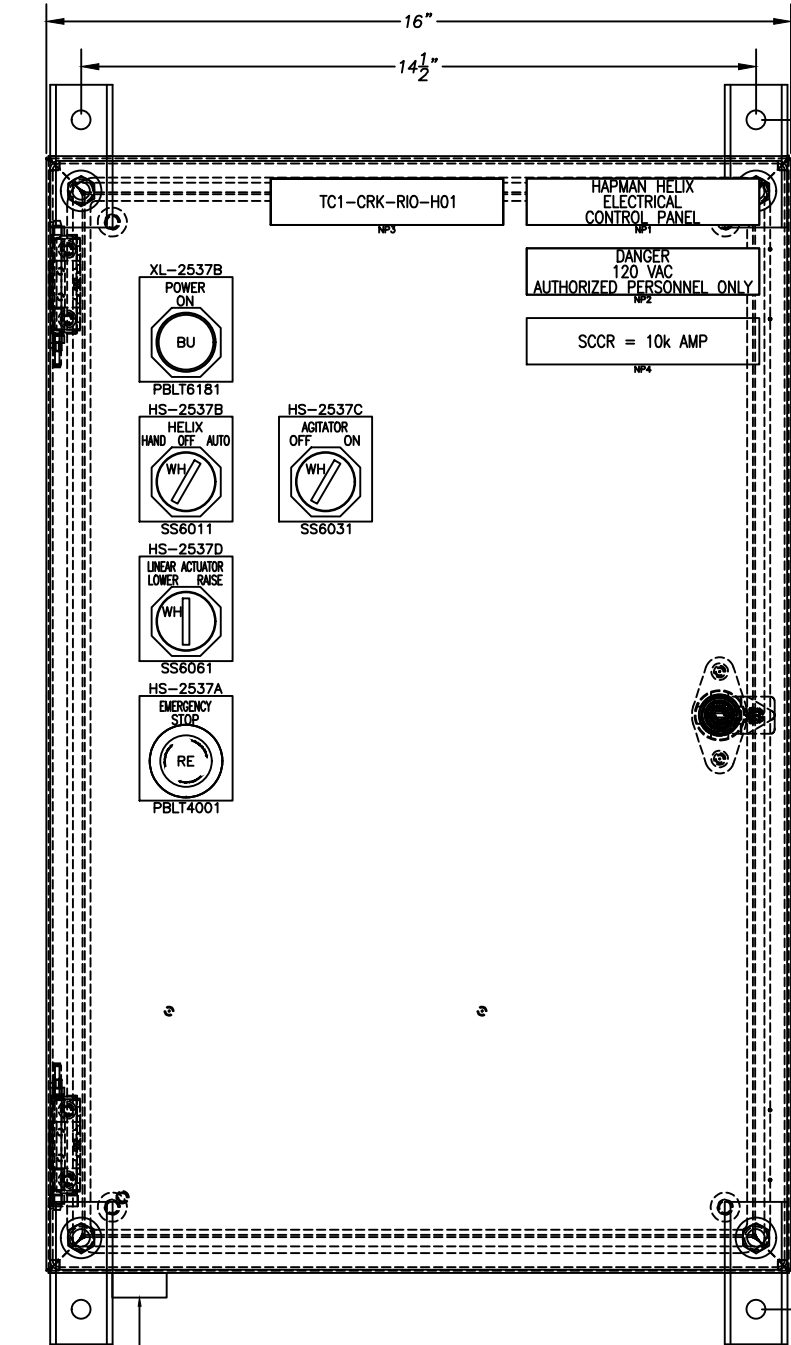
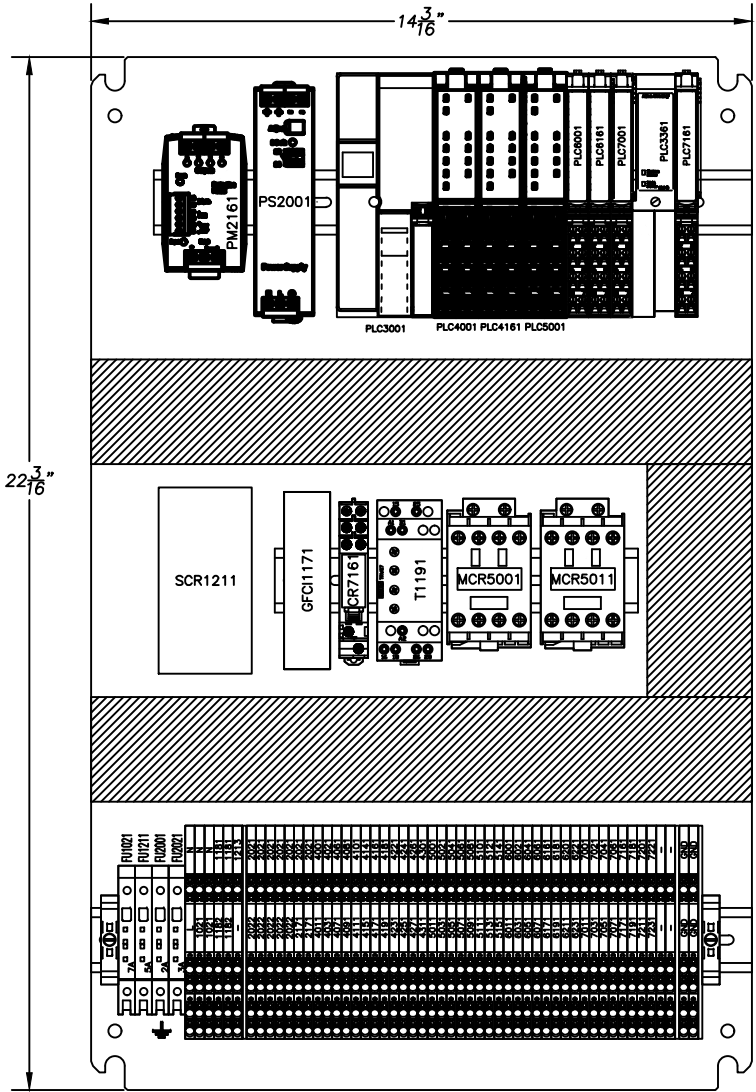
----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING



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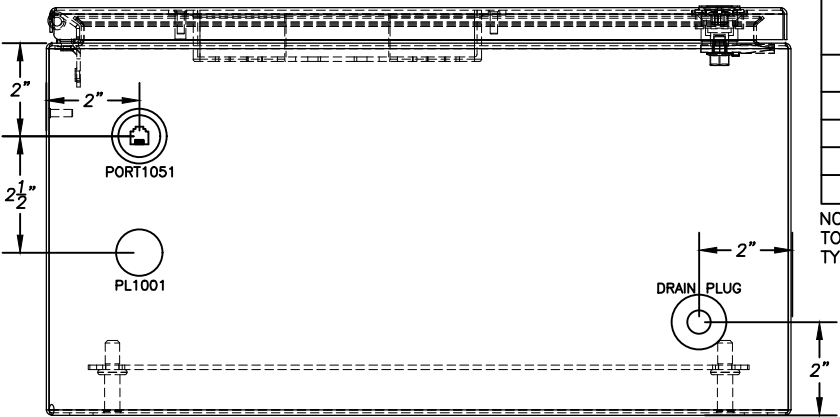
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|--------------------------------------------|----------|----------|--------------|---------|---------|----------|
| DATE                                       | 4/21/23  | JOB NO.  | X20335-AC    |         | DRAWN   | MRL      |
| SCALE                                      | N/A      |          |              | BOM = 1 | DWG = 1 |          |
| TITLE                                      |          |          |              |         |         |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |          |          |              |         |         |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |          |          |              |         |         |          |
| SHEET                                      | 12 OF 16 | DWG. NO. | X20335AC-B00 |         |         | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



| NAMEPLATE DESCRIPTIONS |                 |            |                           |
|------------------------|-----------------|------------|---------------------------|
| N.P. #                 | LINE 1          | LINE 2     | LINE 3                    |
| NP1                    | HAPMAN HELIX    | ELECTRICAL | CONTROL PANEL             |
| NP2                    | DANGER          | 120 VAC    | AUTHORIZED PERSONNEL ONLY |
| NP3                    | TC1-CRK-RIO-H01 |            |                           |
| NP4                    | SCCR = 10k AMPS |            |                           |

- NOTES:
- NP1 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - NP2 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" RED FRONT WITH 1/4" HIGH ENGRAVED WHITE LETTERS.
  - NP3-NP4 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - ALL OTHER NAMEPLATES SHALL BE 3/4"x2" WHITE FRONT WITH BLACK LETTERS



PIN/PLUG CONNECTIONS BY OTHERS

| FUSE REPLACEMENT |           |          |               |
|------------------|-----------|----------|---------------|
| FUSE #           | TYPE/SIZE | AMPERAGE | VOLTAGE       |
| FU1021           | MDL-7-R   | 7A       | 250VAC        |
| FU1211           | MDL-5-R   | 5A       | 250VAC        |
| FU2001           | MDL-2-R   | 2A       | 250VAC        |
| FU2021           | ABC-3-R   | 3A       | 250VAC/125VDC |

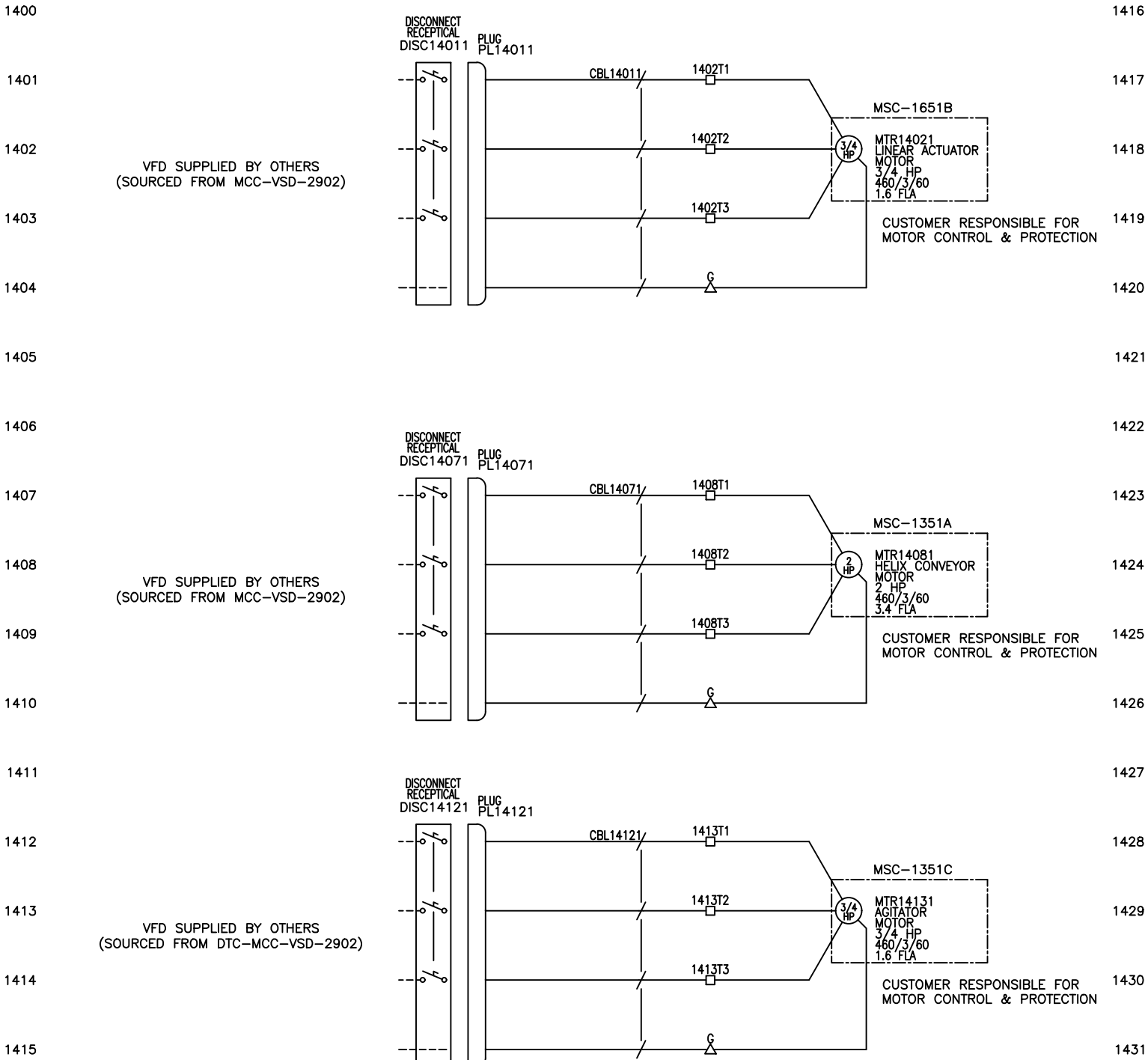
NOTE:  
TO REDUCE THE RISK OF FIRE - REPLACE ONLY WITH THE SAME TYPE AND RATING OF FUSE IN THE FUSE REPLACEMENT CHART.

|                    |                                  |
|--------------------|----------------------------------|
| VOLTAGE:           | 120 VOLT                         |
| PHASE & FREQ.:     | 1PH., 60HZ                       |
| FULL LOAD CURRENT: | 2 AMP                            |
| LARGEST MOTOR FLA: | 1 AMP                            |
| S.C.C.R.:          | 10k AMP                          |
| ENCLOSURE RATING:  | TYPE 4X                          |
| SCHEMATIC:         | X20335AC-B00 (MAIN PANEL)        |
| MFG. BY:           | KMC Global Controls & Automation |
| UL FILE NUMBER:    | E500032                          |

|                                            |                                                                     |                                                                                      |                |
|--------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------|
| <b>HAPMAN</b><br>Ideas that move           |                                                                     | 5944 E. N AVE., KALAMAZOO, MICHIGAN 49048<br>PHONE (269) 343-1675 FAX (269) 349-2477 |                |
| DATE<br>4/21/23                            | JOB NO.<br>X20335-AC                                                | DRAWN<br>BOM = 1                                                                     | MRL<br>DWG = 1 |
| SCALE<br>N/A                               | TITLE<br><b>MAIN PANEL LAYOUT</b><br>TYPE 4X - HELIX CONVEYOR HI/LO |                                                                                      |                |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |                                                                     | REV.<br>0.0                                                                          |                |
| SHEET<br>13 OF 16                          | DWG. NO.<br>X20335AC-B00                                            |                                                                                      |                |



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WIRE GAUGE

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CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
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LEGEND

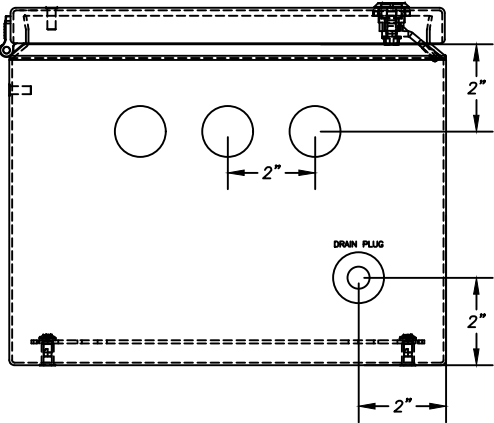
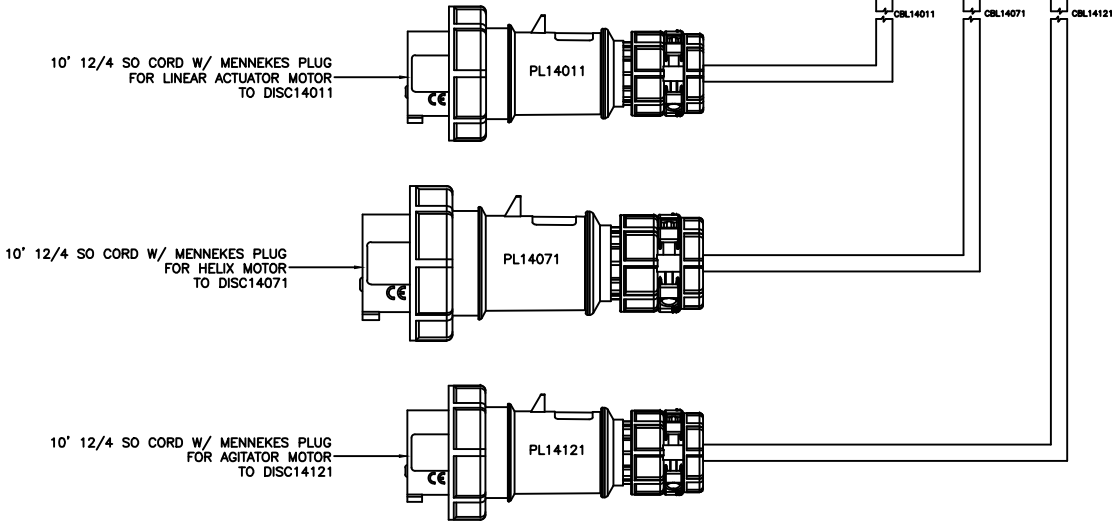
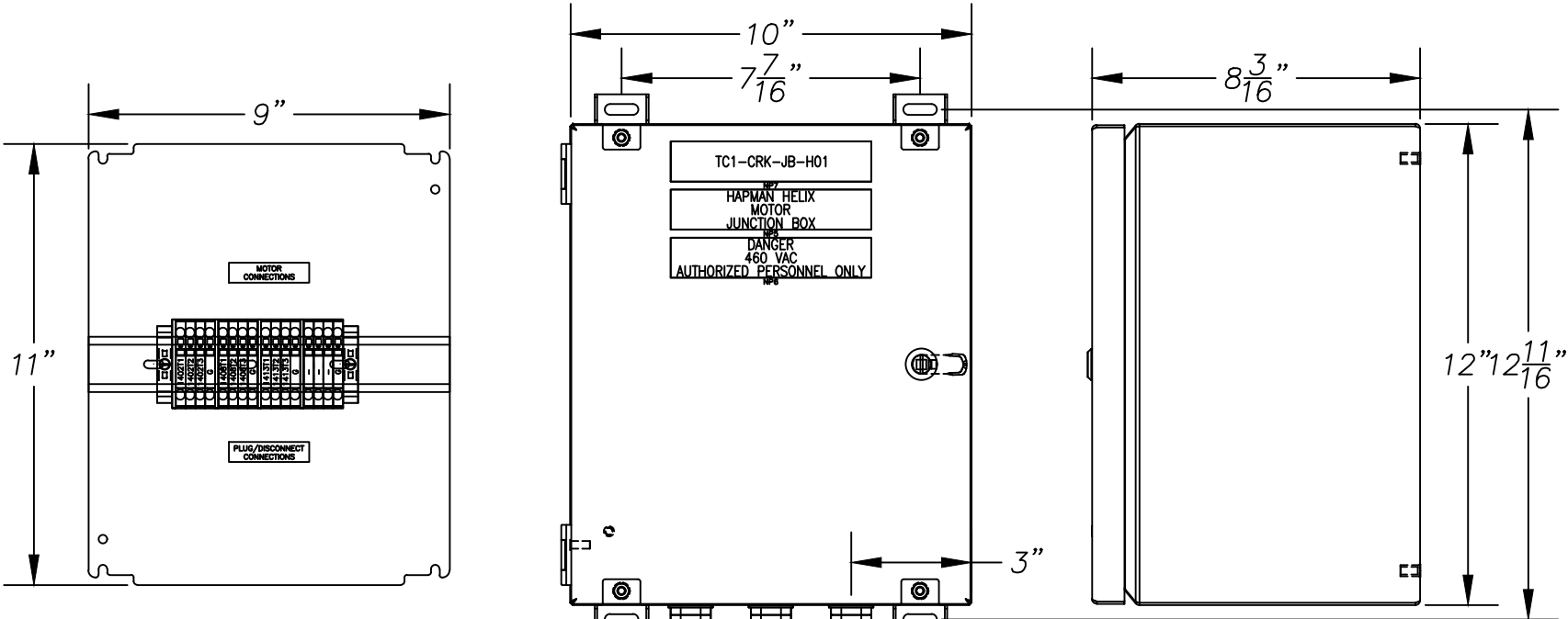
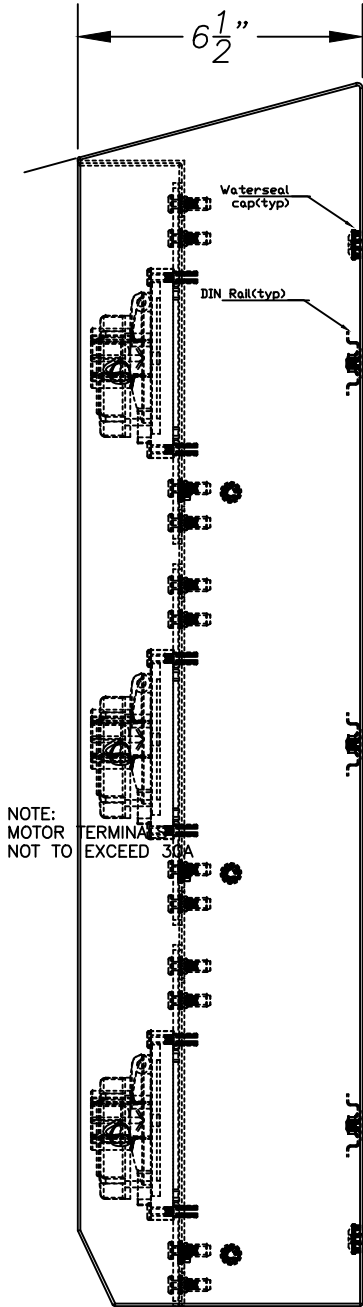
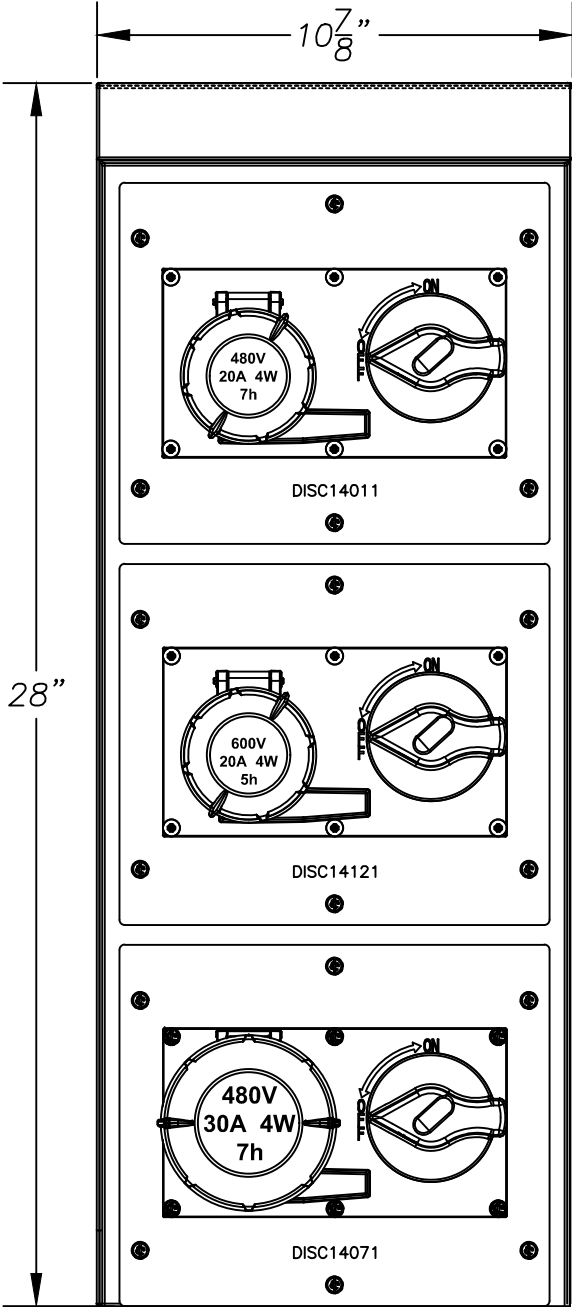
----- FIELD WIRING  
————— PANEL WIRING

**HAPMAN**  
Ideas that move

5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 349-2477

|                                            |         |         |           |          |              |
|--------------------------------------------|---------|---------|-----------|----------|--------------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-AC | DRAWN    | MRL          |
| SCALE                                      | N/A     | BOM     | = 1       | DWG      | = 1          |
| TITLE                                      |         |         |           |          |              |
| MOTOR JUNCTION BOX WIRING DIAGRAM          |         |         |           |          |              |
| TYPE 4X - HELIX CONVEYOR HI/LO             |         |         |           |          |              |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |
| SHEET                                      | 14      | OF      | 16        | DWG. NO. | X20335AC-B00 |
| REV.                                       |         |         |           |          | 0.0          |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



| NAMEPLATE DESCRIPTIONS |                |         |                           |
|------------------------|----------------|---------|---------------------------|
| N.P. #                 | LINE 1         | LINE 2  | LINE 3                    |
| NP5                    | HAPMAN HELIX   | MOTOR   | JUNCTION BOX              |
| NP6                    | DANGER         | 460 VAC | AUTHORIZED PERSONNEL ONLY |
| NP7                    | TC1-CRK-JB-H01 |         |                           |

- NOTES:
- NP5 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - NP6 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" RED FRONT WITH 1/4" HIGH ENGRAVED WHITE LETTERS.
  - NP7 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - ALL OTHER NAMEPLATES SHALL BE 3/4"x2" WHITE FRONT WITH BLACK LETTERS

| (MOTOR JUNCTION BOX) |                                  |
|----------------------|----------------------------------|
| VOLTAGE:             | 460 VOLT                         |
| PHASE & FREQ.:       | 3PH., 60HZ                       |
| ENCLOSURE RATING:    | TYPE 4X                          |
| SCHEMATIC:           | X20335AC-B00                     |
| MFG. BY:             | KMC Global Controls & Automation |
| UL FILE NUMBER       | E500032                          |

|                                                                                      |          |          |              |
|--------------------------------------------------------------------------------------|----------|----------|--------------|
| <b>HAPMAN</b><br><i>ideas that move</i>                                              |          |          |              |
| 5944 E. N AVE., KALAMAZOO, MICHIGAN 49048<br>PHONE (269) 343-1675 FAX (269) 349-2477 |          |          |              |
| DATE                                                                                 | 4/21/23  | JOB NO.  | X20335-AC    |
| SCALE                                                                                | N/A      | DRAWN    | MRL          |
|                                                                                      |          | BOM = 1  | DWG = 1      |
| TITLE                                                                                |          |          |              |
| MOTOR JUNCTION BOX<br>TYPE 4X - HELIX CONVEYOR HI/LO                                 |          |          |              |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                                           |          |          |              |
| SHEET                                                                                | 15 OF 16 | DWG. NO. | X20335AC-B00 |
|                                                                                      |          |          | REV. 0.0     |



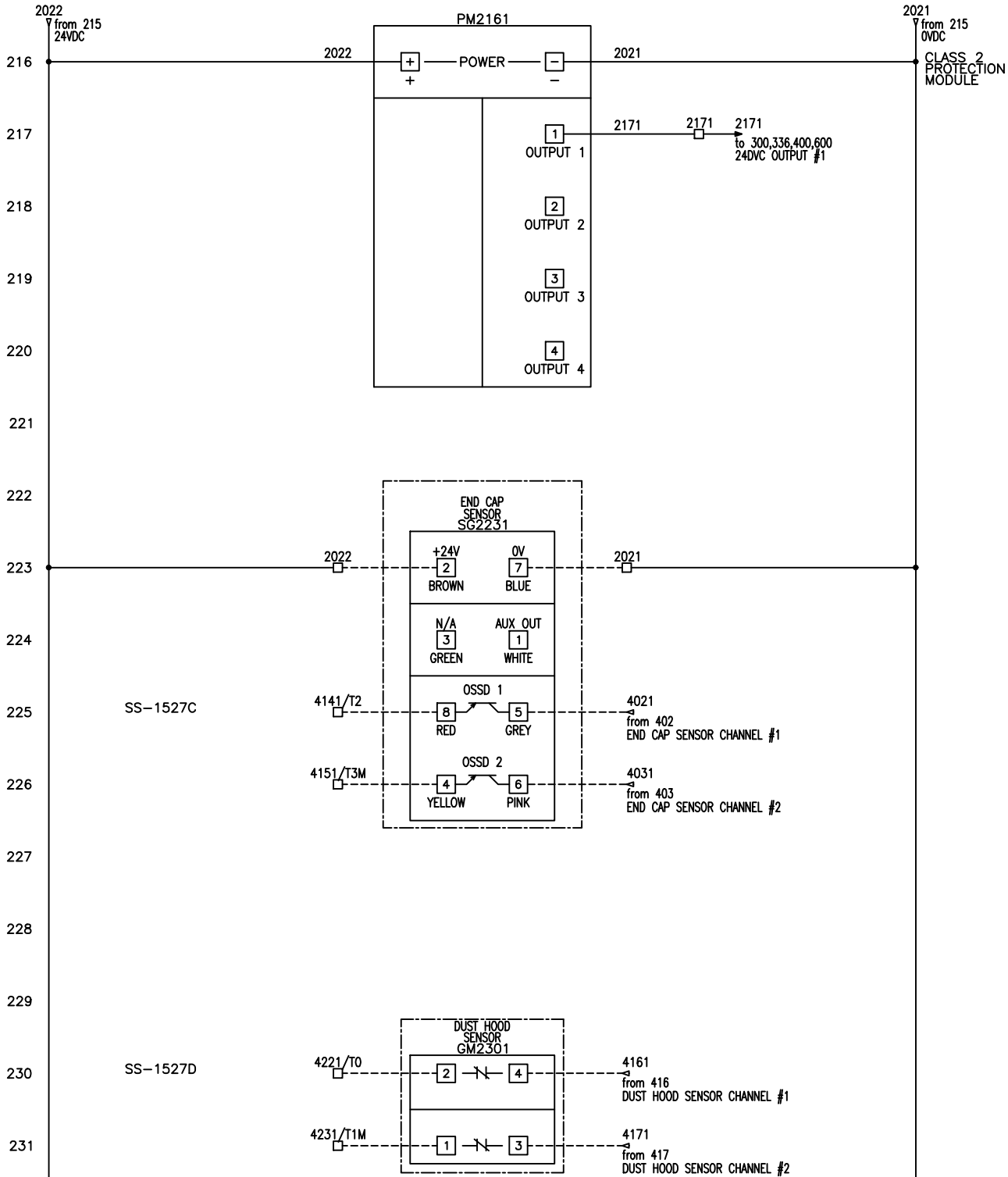
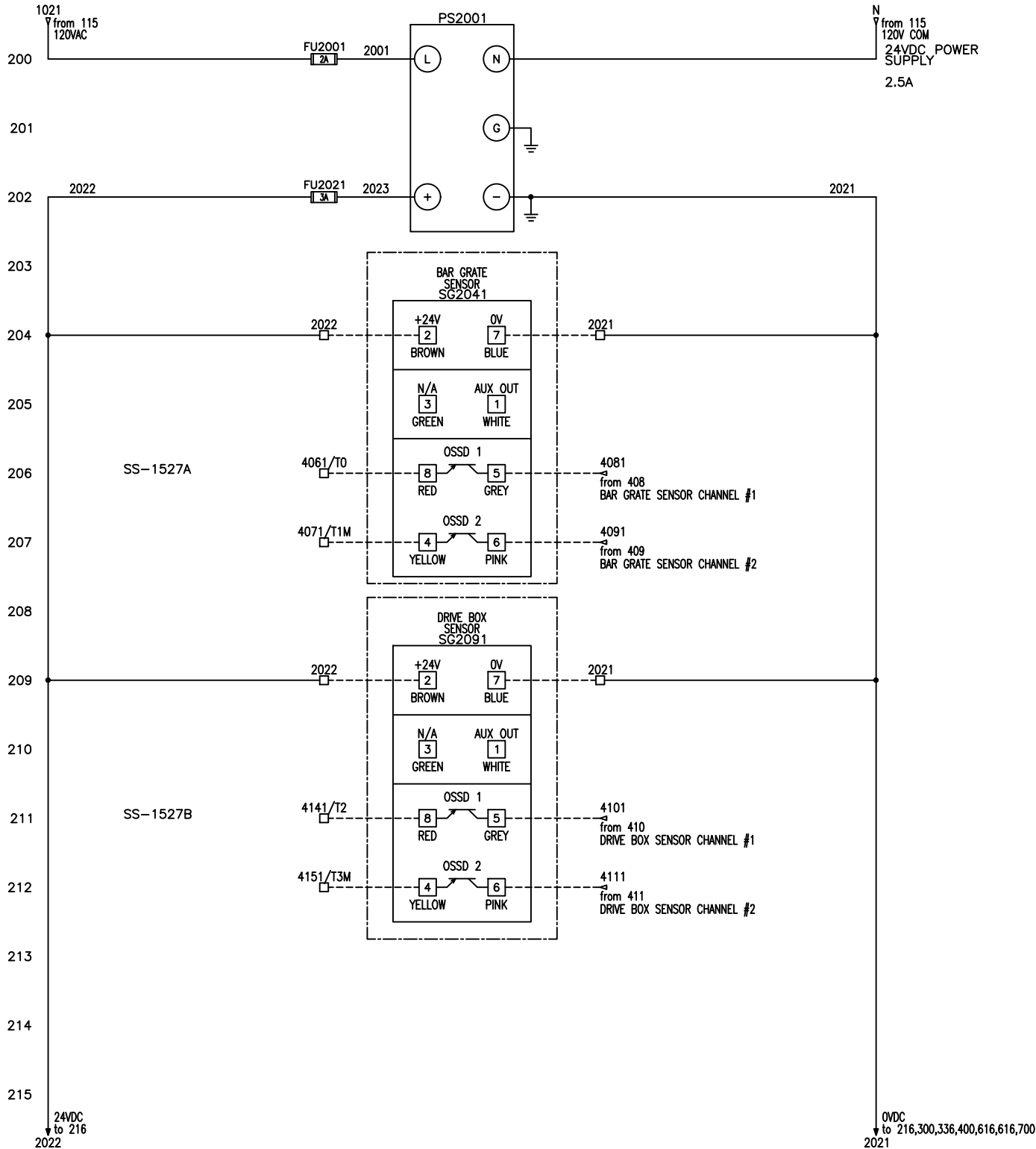
| ITEM | QTY | PART #         | MANUFACTURE  | DESCRIPTION                                                            | DESCRIPTOR                |
|------|-----|----------------|--------------|------------------------------------------------------------------------|---------------------------|
| 1    | 1   | CSD24168SS     | HOFFMAN      | 24"x16"x8" TYPE 4X STAINLESS STEEL ENCLOSURE                           | MAIN PANEL                |
| 2    | 1   | CP2416         | HOFFMAN      | SUB PANEL                                                              | .                         |
| 3    | 1   | CMFKSS         | HOFFMAN      | TYPE 4X STAINLESS STEEL FOOT KIT                                       | .                         |
| 4    | 1   | IAEBHC6        | PANDUIT      | PROG. PORT, RJ45 CAT6                                                  | PORT1051                  |
| 5    | 1   | 576-S10-004    | QUIKTRON     | 4' SHIELDED CAT6 ETHERNET CABLE                                        | ENET-3041                 |
| 6    | 4   | 1492-H6        | AB           | 300V 12A FUSIBLE TERMINAL BLOCK #30-#12AWG                             | FU1021,1211,2001,2021     |
| 7    | 1   | 1492-N37       | AB           | FUSIBLE TERMINAL END BARRIER                                           | FU1021-2021               |
| 8    | 1   | MDL-7-R        | BUSSMANN     | 7A 1/4"x1-1/4" TIME-DELAY GLASS FUSE                                   | FU1021                    |
| 9    | 1   | MDL-5-R        | BUSSMANN     | 5A 1/4"x1-1/4" TIME-DELAY GLASS FUSE                                   | FU1211                    |
| 10   | 1   | MDL-2-R        | BUSSMANN     | 2A 1/4"x1-1/4" TIME-DELAY GLASS FUSE                                   | FU2001                    |
| 11   | 1   | ABC-3-R        | BUSSMANN     | 3A 1/4"x1-1/4" FAST-ACTING CERAMIC FUSE                                | FU2021                    |
| 12   | 1   | 1492-MCGAT115  | AB           | 15A 1 POLE GFCI CIRCUIT BREAKER                                        | GFCI1171                  |
| 13   | 1   | 1492-AMCDIN1   | AB           | GFCI CIRCUIT BREAKER DIN CLIP                                          | GFCI1171                  |
| 14   | 1   | TRW27          | EATON        | 24-240 VAC/DC MULTI-FUNCTION DPDT TIMER                                | T1191                     |
| 15   | 1   | 179502-A       | FMC          | 115/60/1 SCR                                                           | SCR1211                   |
| 16   | 1   | 4SSL1/2DR      | APPLETON     | 2"x4"x2-1/8" HANDYBOX                                                  | SCR1211                   |
| 17   | 1   | 180A           | APPLETON     | 4-1/8"x2-3/8" HANDYBOX COVER                                           | SCR1211                   |
| 18   | 1   | PIM60.245      | PULS         | 24VDC 2.5A NEC CLASS2 COMPLAINT POWER SUPPLY                           | PS2001                    |
| 19   | 1   | PISA11.CLASS2  | PULS         | 24VDC 14.8A PROTECTION MODULE                                          | PM2161                    |
| 20   | 1   | 1734-AENT      | AB           | POINT I/O ETHERNET ADAPTER                                             | PLC3001                   |
| 21   | 1   | 1734-FPD       | AB           | POINT I/O FIELD POTENTIAL DISTRIBUTOR MODULE                           | PLC3361                   |
| 22   | 1   | 1734-IB8S      | AB           | 8 POINT SAFETY SINKING INPUT CARD                                      | PLC4001                   |
| 23   | 1   | 1734-OB8S      | AB           | 8 POINT SAFETY SOURCING OUTPUT CARD                                    | PLC5001                   |
| 24   | 2   | 1734-IB8       | AB           | POINT I/O 8PT DIGITAL INPUT CARD                                       | PLC6001,6161              |
| 25   | 2   | 1734-OB8       | AB           | POINT I/O 8PT DIGITAL OUTPUT CARD                                      | PLC7001,7161              |
| 26   | 10  | 1734-TBS       | AB           | POINT I/O SPRING TERMINAL BASE                                         | PLC4001-7161              |
| 27   | 1   | 800H-FRXTQH2RA | AB           | 30mm 12-130VAC/DC RED ILLUMINATED PUSH/PULL MUSH HEAD PB               | PBLT4001                  |
| 28   | 1   | 800T-XA4       | AB           | 30mm 2 N.C. CONTACT BLOCK                                              | PBLT4001                  |
| 29   | 5   | CT-SPECIAL     | CT           | 30mm LEGEND CUSTOM TEXT (SEE DRAWING)                                  | PBLT4001,6181,SS6011-6181 |
| 30   | 2   | 700-CF400EJ    | AB           | 24VDC CONTROL RELAY WITH 4 N.O. CONTACTS                               | MCR5001,5011              |
| 31   | 1   | 800H-JR2A      | AB           | 30mm 3 POSITION MAINTAINED SELECTOR SWITCH                             | SS6011                    |
| 32   | 1   | 800H-HR2D1     | AB           | 30mm 2 POSITION MAINTAINED SELECTOR SWITCH                             | SS6031                    |
| 33   | 1   | 800H-JR91A     | AB           | 3 POS. SPRING RETURN SELECTOR SWITCH W/ 1 N.O. & 1 N.C. CONTATCS       | SS6061                    |
| 34   | 1   | 800H-QRBH2B    | AB           | 30mm 12-130VAC/DC BLUE ILLUM. FLUSH HEAD PB W/ 1 N.O. & 1 N.C. CONTACT | PBLT6181                  |
| 35   | 1   | 700-HK32Z24    | AB           | 24VDC DPDT SLIM RELAY CUBE                                             | CR7001                    |
| 36   | 1   | 700-HN222      | AB           | DPDT SLIM RELAY SOCKET                                                 | CR7001                    |
| 37   | 48  | 3210609        | PHOENIX      | 5.2mm DOUBLE STACK PUSH IN TERMINAL                                    | .                         |
| 38   | 2   | 3210611        | PHOENIX      | 5.2mm DOUBLE STACK PUSH IN GROUND TERMINAL                             | .                         |
| 39   | 3   | 3210613        | PHOENIX      | END PLATE FOR 5.2mm DOUBLE STACK PUSH IN TERMINAL                      | .                         |
| 40   | 2   | 0800886        | PHOENIX      | SCREW ON END STOP                                                      | .                         |
| 41   | 2'  | 111023         | E-RAIL       | 35mm DIN RAIL                                                          | .                         |
| 42   | 4'  | F2X4LG6        | PANDUIT      | 2"x4" WIREWAY                                                          | .                         |
| 43   | 4'  | C2LG6          | PANDUIT      | 2" WIREWAY COVER                                                       | .                         |
| 44   | 4   | JPPT4A         | CT           | 1"x5" CUSTOM ENGRAVED LEGEND                                           | NP1-4                     |
| 45   | 1   | SSTC 1CD       | CROUSE-HINDS | DRAIN PULG, 1/2" STAINLESS STEEL                                       | .                         |
| 46   | 1   | 212PA          | MENCOM       | 1/2" NPT LOCKNUT                                                       | .                         |
| 47   | 1   | PPS0305W2100   | PANDUIT      | ARC FLASH LABEL                                                        | .                         |
| 48   |     |                |              |                                                                        |                           |
| 49   |     |                |              |                                                                        |                           |
| 50   |     |                |              |                                                                        |                           |
| 51   |     |                |              |                                                                        |                           |
| 52   |     |                |              |                                                                        |                           |
| 53   |     |                |              |                                                                        |                           |
| 54   |     |                |              |                                                                        |                           |
| 55   |     |                |              |                                                                        |                           |
| 56   |     |                |              |                                                                        |                           |

| ITEM | QTY | PART #         | MANUFACTURE   | DESCRIPTION                                                | DESCRIPTOR             |
|------|-----|----------------|---------------|------------------------------------------------------------|------------------------|
| 57   | 1   | 888N-M4AF1-1F  | AB            | 4 PIN 600V, 10A, 16 AWG MALE RECEPTACLE 1' LEAD            | PL1001                 |
| 58   | 1   | 889N-F4AFC-12F | AB            | 4 PIN 600V, 10A, FEMALE QUICK CHANGE CONNECTOR W/ 12' LEAD | RECEP1001 (SHIP LOOSE) |
| 59   | REF | 855T-B24TL4    | ALLEN BRADLEY | 24VDC STEADY LED TOWER LIGHT- (AB), RED                    | LT7011                 |
| 60   | REF | 855T-BCBC      | ALLEN BRADLEY | SURFACE MOUNT BASE- (AB), W/CAP, 1/2 NPT                   | LT7011                 |
| 61   |     |                |               |                                                            |                        |
| 62   |     |                |               |                                                            |                        |
| 63   | 1   | 12108ELJSS     | SCE           | 12"x10"x8" TYPE 4X STAINLESS STEEL ENCLOSURE               | MOTOR JUNCTION BOX     |
| 64   | 1   | 12P10          | SCE           | SUB PANEL                                                  | .                      |
| 65   | 12  | 3211757        | PHOENIX       | 6.2mm 30A PUSH IN TERMINAL                                 | .                      |
| 66   | 4   | 3211766        | PHOENIX       | 6.2mm PUSH IN GROUND TERMINAL                              | .                      |
| 67   | 4   | 3030420        | PHOENIX       | END COVER FOR 6.2mm PUSH IN TERMINAL                       | .                      |
| 68   | 2   | 0800886        | PHOENIX       | SCREW ON END STOP                                          | .                      |
| 69   | 1'  | 111023         | E-RAIL        | 35mm DIN RAIL                                              | .                      |
| 70   | 3   | SHC1036        | HUBBELL       | 3/4" NPT MACHINED ALUMINUM CORD GRIP (0.5-0.63")           | CBL14011,14071,14121   |
| 71   | 3   | 20509002       | HUBBELL       | 3/4" NPT METAL CLAD SEALING O-RING                         | CBL14011,14071,14121   |
| 72   | 3   | 234PA          | MENCOM        | 3/4" NPT LOCKNUT                                           | CBL14011,14071,14121   |
| 73   | 33' | 12/4 SO CORD   | CAROL         | 12/4 SO CORD                                               | CBL14011,14071,14121   |
| 74   | 1   | ME420P7W       | MENNEKES      | 20A 3P 4W 480V WATERTIGHT PLUG                             | PL14011                |
| 75   | 1   | ME430P7W       | MENNEKES      | 30A 3P 4W 480V WATERTIGHT PLUG                             | PL14071                |
| 76   | 1   | ME420P5W       | MENNEKES      | 20A 3P 4W 600V WATERTIGHT PLUG                             | PL14121                |
| 77   | 1   | SPX3A-639068   | MENNEKES      | AMAXX STAINLESS STEEL 3 POSITION INTERLOCKED RECEPTACLE    | DISC14011,14071,14121  |
| 78   | 3   | JPPT4A         | CT            | 1"x5" CUSTOM ENGRAVED LEGEND                               | NP5-7                  |
| 79   | 1   | SSTC 1CD       | CROUSE-HINDS  | DRAIN PULG, 1/2" STAINLESS STEEL                           | .                      |
| 80   | 1   | 212PA          | MENCOM        | 1/2" NPT LOCKNUT                                           | .                      |
| 81   | 1   | PPS0305W2100   | PANDUIT       | ARC FLASH LABEL                                            | .                      |
| 82   | 1   | 31622003L      | HUBBELL       | 1/2" NPT LOCKING NUT                                       | .                      |
| 83   |     |                |               |                                                            |                        |
| 84   |     |                |               |                                                            |                        |
| 85   |     |                |               |                                                            |                        |
| 86   |     |                |               |                                                            |                        |
| 87   |     |                |               |                                                            |                        |
| 88   |     |                |               |                                                            |                        |
| 89   |     |                |               |                                                            |                        |
| 90   |     |                |               |                                                            |                        |
| 91   |     |                |               |                                                            |                        |
| 92   |     |                |               |                                                            |                        |
| 93   |     |                |               |                                                            |                        |
| 94   |     |                |               |                                                            |                        |
| 95   |     |                |               |                                                            |                        |
| 96   |     |                |               |                                                            |                        |
| 97   |     |                |               |                                                            |                        |
| 98   |     |                |               |                                                            |                        |
| 99   |     |                |               |                                                            |                        |
| 100  |     |                |               |                                                            |                        |
| 101  |     |                |               |                                                            |                        |
| 102  |     |                |               |                                                            |                        |
| 103  |     |                |               |                                                            |                        |
| 104  |     |                |               |                                                            |                        |
| 105  |     |                |               |                                                            |                        |

|                                                                     |                          |                                                                                      |                |
|---------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------|----------------|
| <b>HAPMAN</b><br><small>ideas that move</small>                     |                          | 5944 E. N AVE., KALAMAZOO, MICHIGAN 49048<br>PHONE (269) 343-1675 FAX (269) 349-2477 |                |
| DATE<br>4/21/23                                                     | JOB NO.<br>X20335-AC     | DRAWN<br>BOM = 1                                                                     | MRL<br>DWG = 1 |
| TITLE<br><b>BILL OF MATERIALS</b><br>TYPE 4X - HELIX CONVEYOR HI/LO |                          |                                                                                      |                |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                          |                          |                                                                                      |                |
| SHEET<br>16 OF 16                                                   | DWG. NO.<br>X20335AC-B00 | REV.<br>0.0                                                                          |                |



| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



CONDUCTOR WIRE CODE

BLACK: GREATER THAN 120VAC  
RED: 120VAC OR LESS  
WHITE: GROUNDED AC LINE  
BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
RETURN/SUPPLY: SHALL BE EQUAL AWG

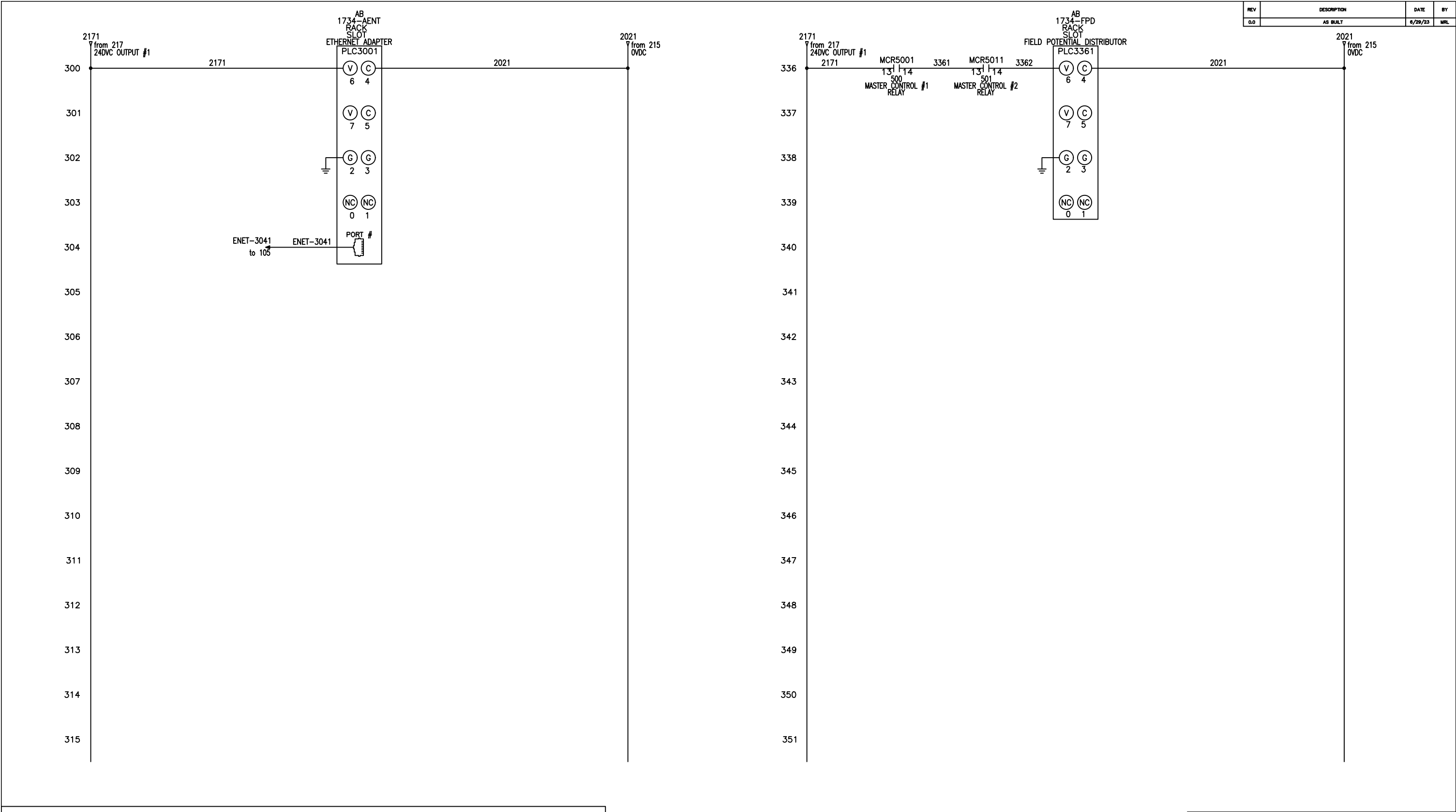
LEGEND

----- FIELD WIRING  
----- PANEL WIRING

**HAPMAN**  
ideas that move

5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 349-2477

|                                                        |         |         |           |          |              |
|--------------------------------------------------------|---------|---------|-----------|----------|--------------|
| DATE                                                   | 4/21/23 | JOB NO. | X20335-BC | DRAWN    | MRL          |
| SCALE                                                  | N/A     |         |           | BOM = 1  | DWG = 1      |
| TITLE                                                  |         |         |           |          |              |
| 24VDC WIRING DIAGRAM<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)             |         |         |           |          |              |
| SHEET                                                  | 2       | OF      | 17        | DWG. NO. | X20335BC-B00 |
|                                                        |         |         |           |          | REV. 0.0     |



336

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348

349

350

351

1734-FPD

RACK

SLOT

FIELD POTENTIAL DISTRIBUTOR

PLC3361

V

C

6

4

V

C

7

5

G

G

2

3

NC

NC

0

1

13

14

500

MASTER CONTROL #1

RELAY

13

14

501

MASTER CONTROL #2

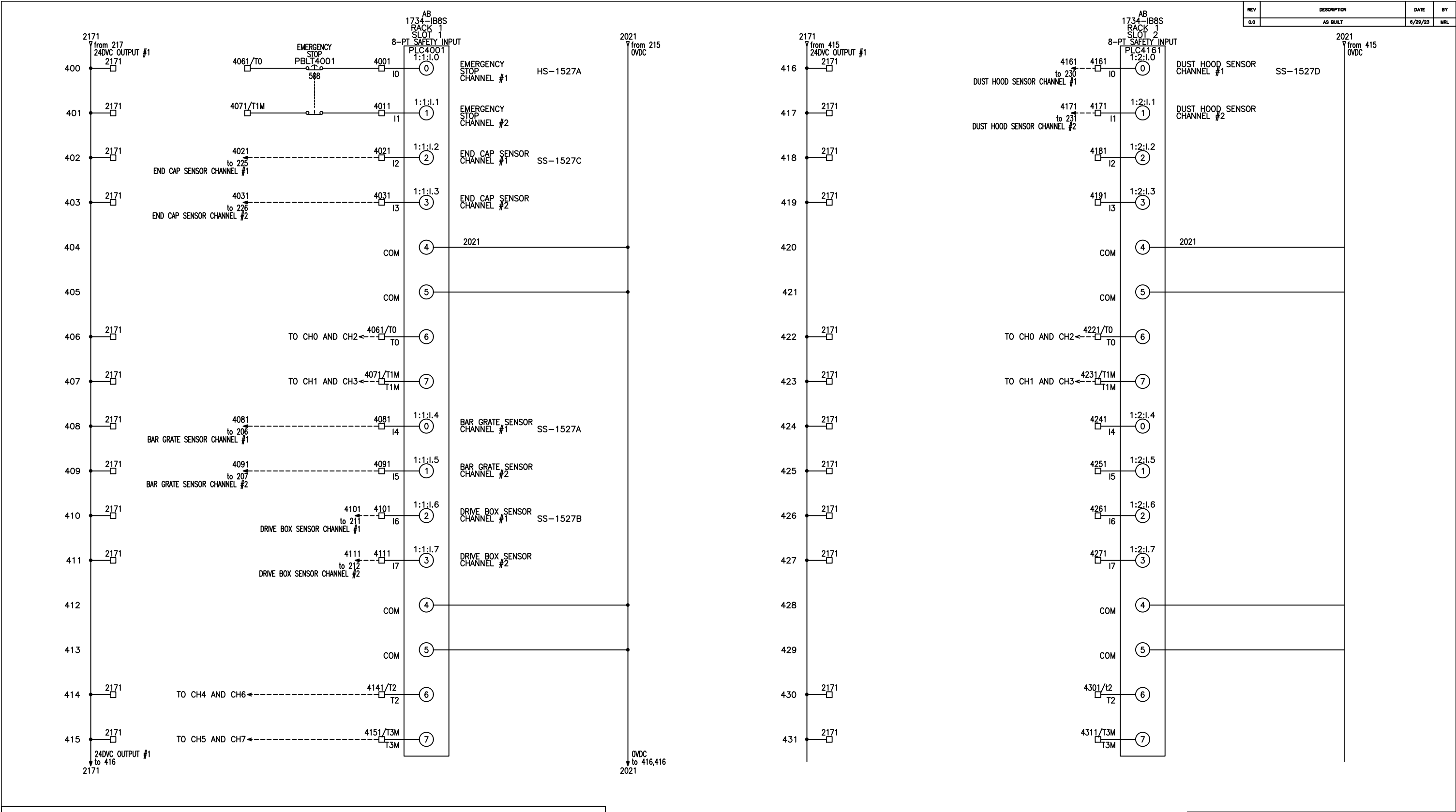
RELAY

| <u>CONDUCTOR WIRE CODE</u> |                                | <u>WIRE GAUGE</u> | <u>LEGEND</u> |                    |  |
|----------------------------|--------------------------------|-------------------|---------------|--------------------|--|
| BLACK:                     | GREATER THAN 120VAC            | MOTOR CIRCUITS:   | MIN. 12 AWG   | ----- FIELD WIRING |  |
| RED:                       | 120VAC OR LESS                 | CONTROL CIRCUITS: | MIN. 16 AWG   |                    |  |
| WHITE:                     | GROUNDING AC LINE              | PLC I/O:          | MIN. 18 AWG   |                    |  |
| BLUE:                      | UN GROUNDED VDC                | FUSES:            | -----         | ----- PANEL WIRING |  |
| WH/BLU:                    | GROUNDING VDC                  |                   |               |                    |  |
| GREEN:                     | GROUNDING CONDUCTORS           |                   |               |                    |  |
| ORANGE:                    | CIRCUITS THAT REMAIN ENERGIZED | RETURN/SUPPLY:    | -----         |                    |  |
|                            |                                |                   |               |                    |  |
| WHEN MAIN DISC. IS "OFF"   |                                |                   |               |                    |  |
|                            |                                |                   |               |                    |  |

**HAPMAN**  
ideas that move

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|                                            |         |         |           |          |              |      |
|--------------------------------------------|---------|---------|-----------|----------|--------------|------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-BC |          | DRAWN        | MRL  |
| SCALE                                      | N/A     |         | BOM =     | 1        | DWG =        | 1    |
| TITLE                                      |         |         |           |          |              |      |
| PLC WIRING DIAGRAM                         |         |         |           |          |              |      |
| TYPE 4X - HELIX CONVEYOR HI/LO             |         |         |           |          |              |      |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |      |
| SHEET                                      | 3       | OF      | 17        | DWG. NO. | X20335BC-B00 | REV. |
|                                            |         |         |           |          |              | 0.0  |



CONDUCTOR WIRE CODE

BLACK: GREATER THAN 120VAC  
RED: 120VAC OR LESS  
WHITE: GROUNDED AC LINE  
BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL AWG

LEGEND

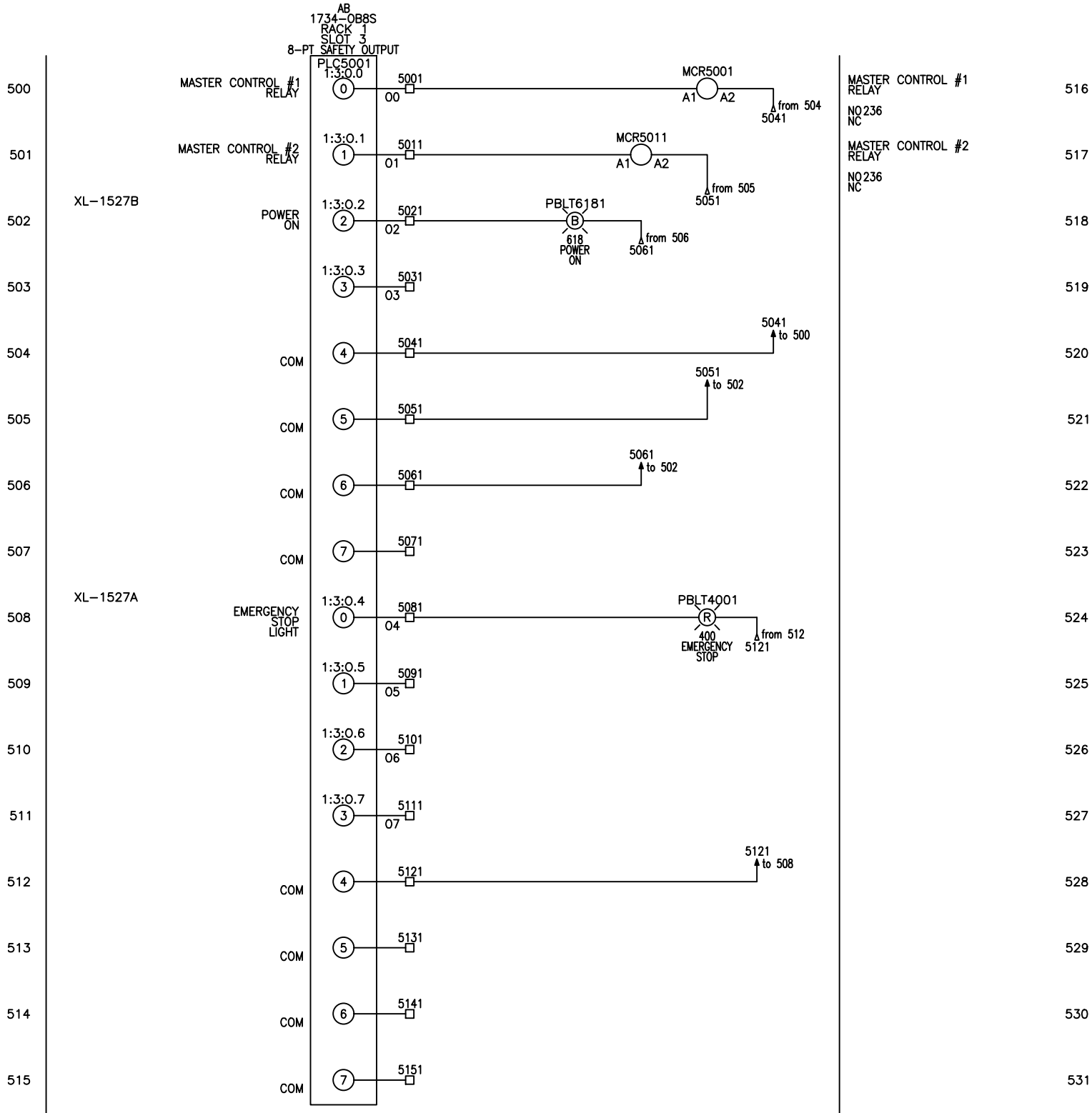
----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING



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|                                            |         |         |           |          |              |
|--------------------------------------------|---------|---------|-----------|----------|--------------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-BC | DRAWN    | MRL          |
| SCALE                                      | N/A     | BOM     | = 1       | DWG      | = 1          |
| TITLE                                      |         |         |           |          |              |
| PLC WIRING DIAGRAM                         |         |         |           |          |              |
| TYPE 4X - HELIX CONVEYOR HI/LO             |         |         |           |          |              |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |
| SHEET                                      | 4       | OF      | 17        | DWG. NO. | X20335BC-B00 |
| REV.                                       |         |         |           |          | 0.0          |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



CONDUCTOR WIRE CODE

BLACK: GREATER THAN 120VAC  
RED: 120VAC OR LESS  
WHITE: GROUNDED AC LINE  
BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
RETURN/SUPPLY: SHALL BE EQUAL AWG

LEGEND

----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING

**HAPMAN**

*ideas that move*

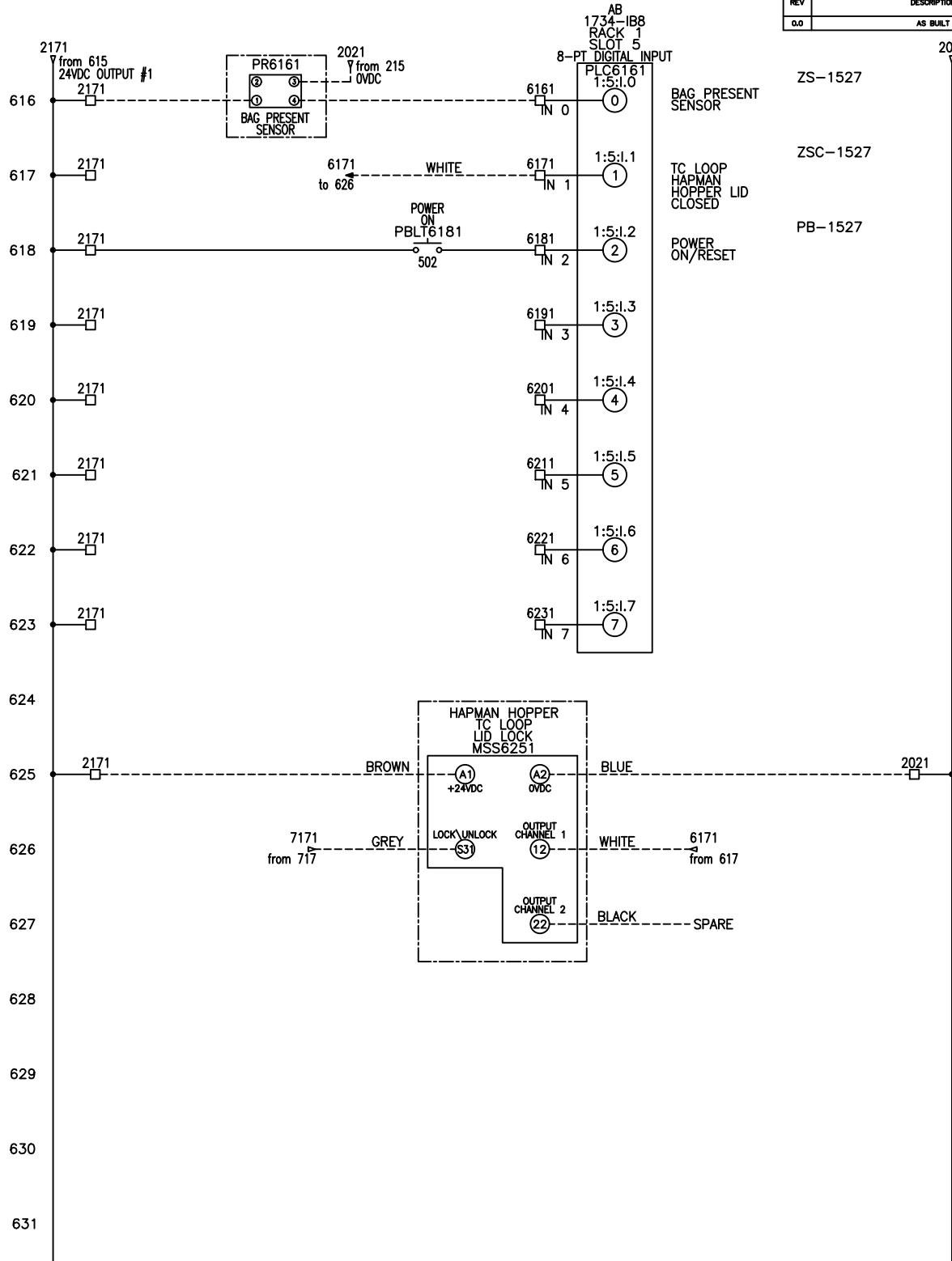
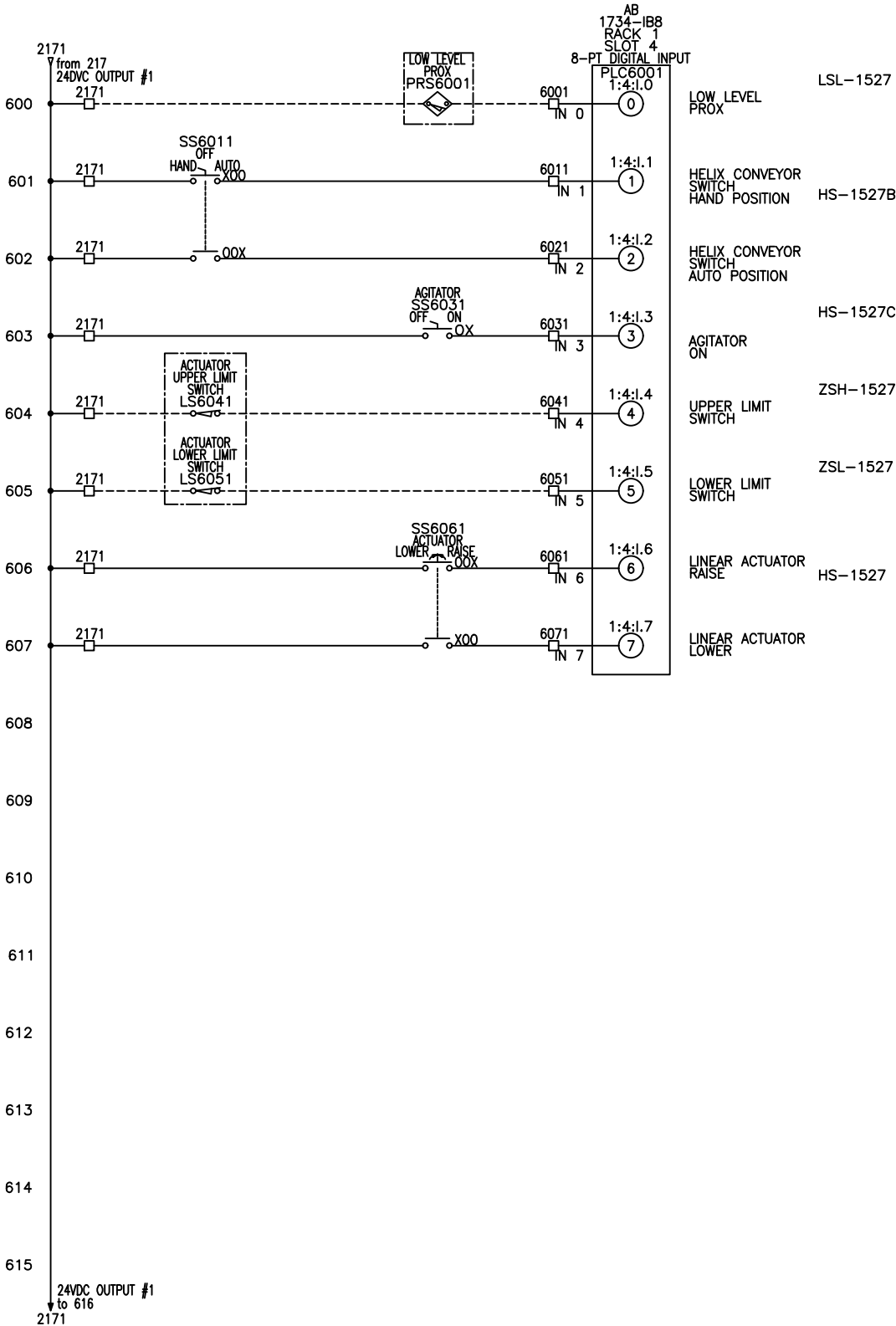
5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 349-2477

|              |                   |         |         |
|--------------|-------------------|---------|---------|
| DATE 4/21/23 | JOB NO. X20335-BC | DRAWN   | MRL     |
| SCALE N/A    |                   | BOM = 1 | DWG = 1 |

TITLE  
**PLC WIRING DIAGRAM**  
TYPE 4X - HELIX CONVEYOR HI/LO

|                                            |                       |
|--------------------------------------------|-----------------------|
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) | REV. 0.0              |
| SHEET 5 OF 17                              | DWG. NO. X20335BC-B00 |





CONDUCTOR WIRE CODE

BLACK: GREATER THAN 120VAC  
RED: 120VAC OR LESS  
WHITE: GROUNDED AC LINE  
BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS:  
CONTROL CIRCUITS:  
PLC I/O:

MIN. 12 AWG  
MIN. 16 AWG  
MIN. 18 AWG

FUSES:

RETURN/SUPPLY:

LINE/LOAD SIDE  
SHALL BE EQUAL  
AWG

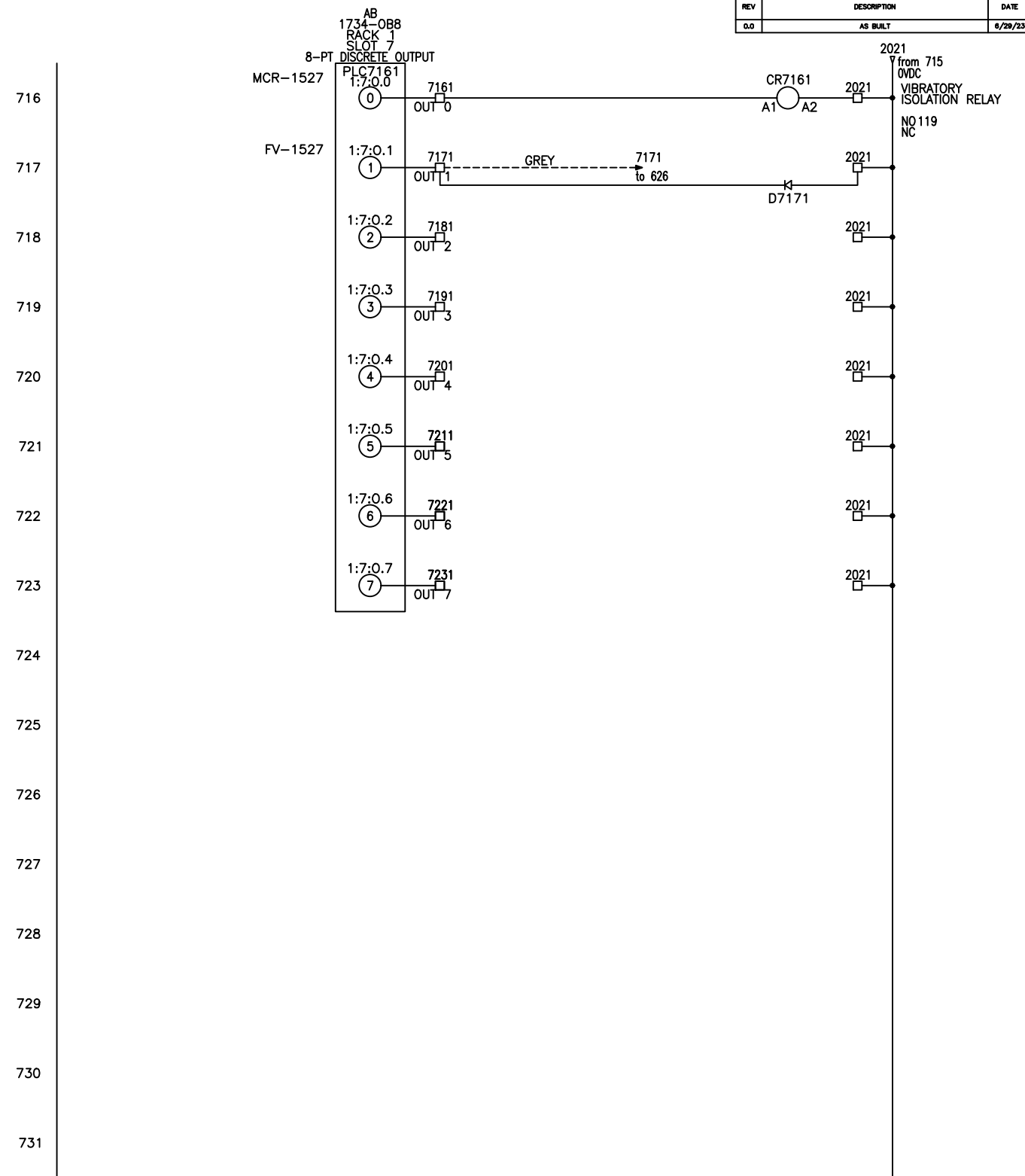
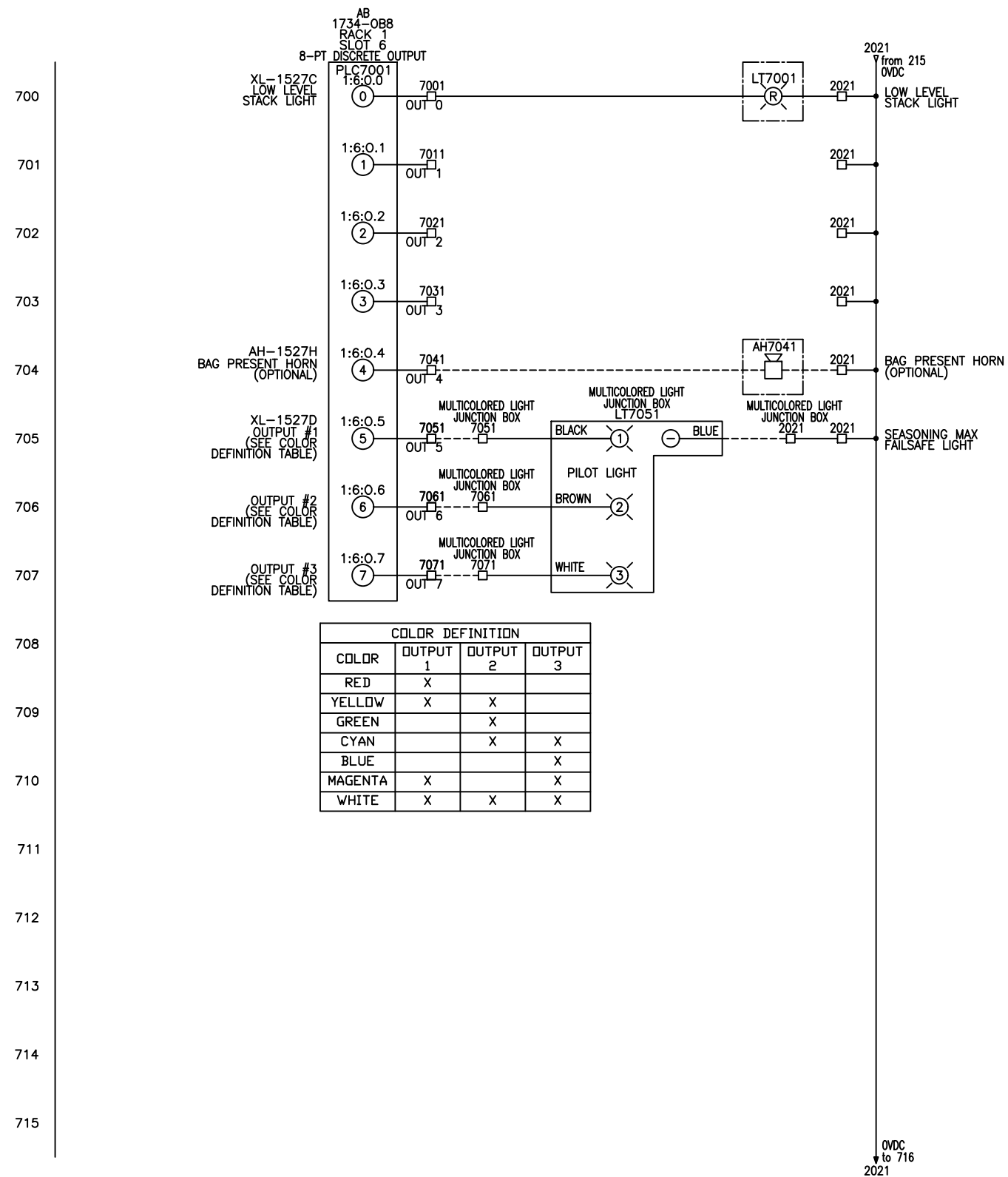
SHALL BE EQUAL  
AWG

LEGEND

----- FIELD WIRING  
----- PANEL WIRING

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

|                                                                      |                      |                                                                                      |                |
|----------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------|----------------|
| <b>HAPMAN</b><br><small>ideas that move</small>                      |                      | 5944 E. N AVE., KALAMAZOO, MICHIGAN 49048<br>PHONE (269) 343-1675 FAX (269) 349-2477 |                |
| DATE<br>4/21/23                                                      | JOB NO.<br>X20335-BC | DRAWN<br>BOM = 1                                                                     | MRL<br>DWG = 1 |
| TITLE<br><b>PLC WIRING DIAGRAM</b><br>TYPE 4X - HELIX CONVEYOR HI/LO |                      |                                                                                      |                |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                           |                      |                                                                                      |                |
| SHEET<br>6                                                           | OF<br>17             | DWG. NO.<br>X20335BC-B00                                                             | REV.<br>0.0    |



**LEGEND**

----- FIELD WIRING

————— PANEL WIRING

|                                                                                       |         |                                                                                       |         |
|---------------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------------------|---------|
|  |         | 5944 E. N. AVE., KALAMAZOO, MICHIGAN 49045<br>PHONE (269) 343-1675 FAX (269) 349-2477 |         |
| DATE                                                                                  | 4/21/23 | JOB NO.                                                                               |         |
| SCALE                                                                                 | N/A     | X20335-BC                                                                             |         |
|                                                                                       |         | DRAWN                                                                                 | MRL     |
|                                                                                       |         | BOM = 1                                                                               | DWG = 1 |
| TITLE                                                                                 |         |                                                                                       |         |
| PLC WIRING DIAGRAM<br>TYPE 4X - HELIX CONVEYOR HI/LO                                  |         |                                                                                       |         |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                                            |         |                                                                                       |         |
| SHEET                                                                                 | 7 OF 17 | DWG. NO.                                                                              |         |
|                                                                                       |         | X20335BC-B00                                                                          |         |
|                                                                                       |         |                                                                                       | REV. 00 |



| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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CONDUCTOR WIRE CODE

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WHITE: GROUNDED AC LINE  
BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED  
WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
  
FUSES: LINE/LOAD SIDE  
SHALL BE EQUAL  
AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL  
AWG

LEGEND

----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING



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|                                            |         |         |           |          |              |      |
|--------------------------------------------|---------|---------|-----------|----------|--------------|------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-BC |          | DRAWN        | MRL  |
| SCALE                                      | N/A     |         | BOM = 1   | DWG = 1  |              |      |
| TITLE                                      |         |         |           |          |              |      |
| RESERVED                                   |         |         |           |          |              |      |
| TYPE 4X - HELIX CONVEYOR HI/LO             |         |         |           |          |              |      |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |      |
| SHEET                                      | 8       | OF      | 17        | DWG. NO. | X20335BC-B00 | REV. |
|                                            |         |         |           |          |              | 0.0  |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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CONDUCTOR WIRE CODE

BLACK: GREATER THAN 120VAC  
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WHITE: GROUNDED AC LINE  
BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL AWG

LEGEND

----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING



5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 349-2477

|                                            |         |         |           |          |              |          |
|--------------------------------------------|---------|---------|-----------|----------|--------------|----------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-BC |          | DRAWN        | MRL      |
| SCALE                                      | N/A     |         | BOM = 1   | DWG = 1  |              |          |
| TITLE                                      |         |         |           |          |              |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |          |
| SHEET                                      | 9       | OF      | 17        | DWG. NO. | X20335BC-B00 | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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CONDUCTOR WIRE CODE

BLACK: GREATER THAN 120VAC  
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ORANGE: CIRCUITS THAT REMAIN ENERGIZED  
WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL AWG

LEGEND

----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING



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PHONE (269) 343-1675 FAX (269) 349-2477

|                                            |         |         |           |          |              |          |
|--------------------------------------------|---------|---------|-----------|----------|--------------|----------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-BC |          | DRAWN        | MRL      |
| SCALE                                      | N/A     |         |           | BOM = 1  | DWG = 1      |          |
| TITLE                                      |         |         |           |          |              |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |          |
| SHEET                                      | 10      | OF      | 17        | DWG. NO. | X20335BC-B00 | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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CONDUCTOR WIRE CODE

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BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
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ORANGE: CIRCUITS THAT REMAIN ENERGIZED WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
  
FUSES: LINE/LOAD SIDE SHALL BE EQUAL AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL AWG

LEGEND

----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING



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PHONE (269) 343-1675 FAX (269) 349-2477

|                                            |         |         |           |          |              |          |
|--------------------------------------------|---------|---------|-----------|----------|--------------|----------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-BC |          | DRAWN        | MRL      |
| SCALE                                      | N/A     |         | BOM = 1   | DWG = 1  |              |          |
| TITLE                                      |         |         |           |          |              |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |          |
| SHEET                                      | 11      | OF      | 17        | DWG. NO. | X20335BC-B00 | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |

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CONDUCTOR WIRE CODE

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BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED  
WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
  
FUSES: LINE/LOAD SIDE  
SHALL BE EQUAL  
AWG  
  
RETURN/SUPPLY: SHALL BE EQUAL  
AWG

LEGEND

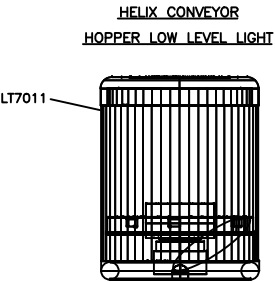
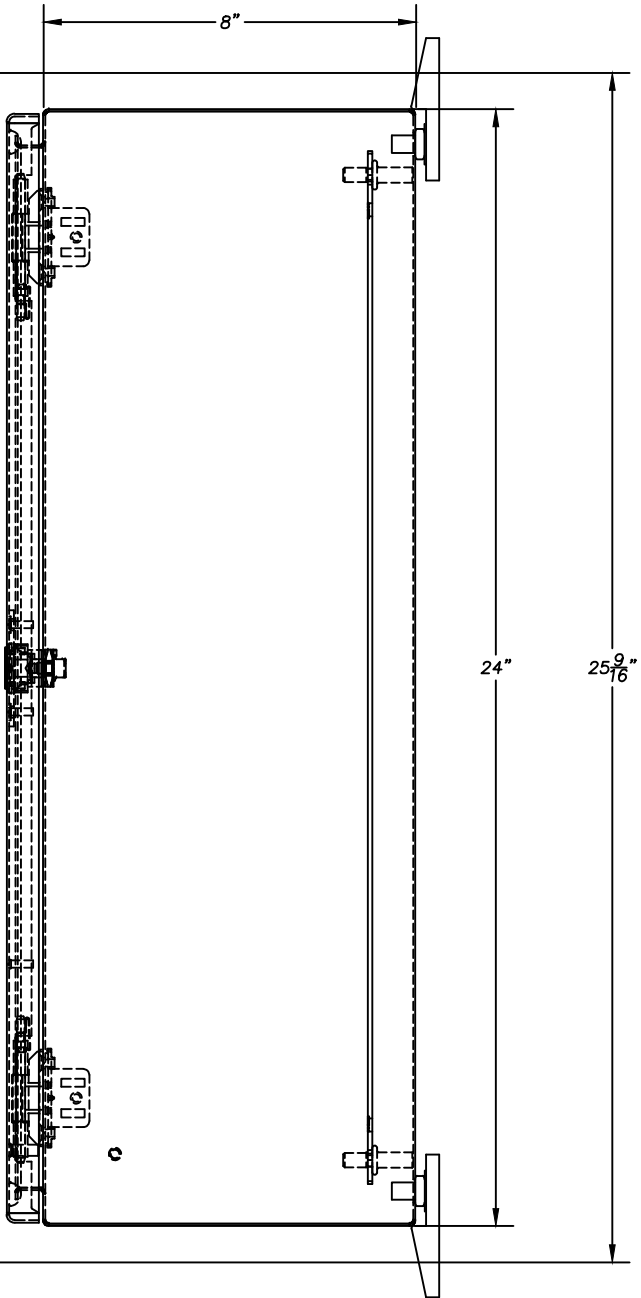
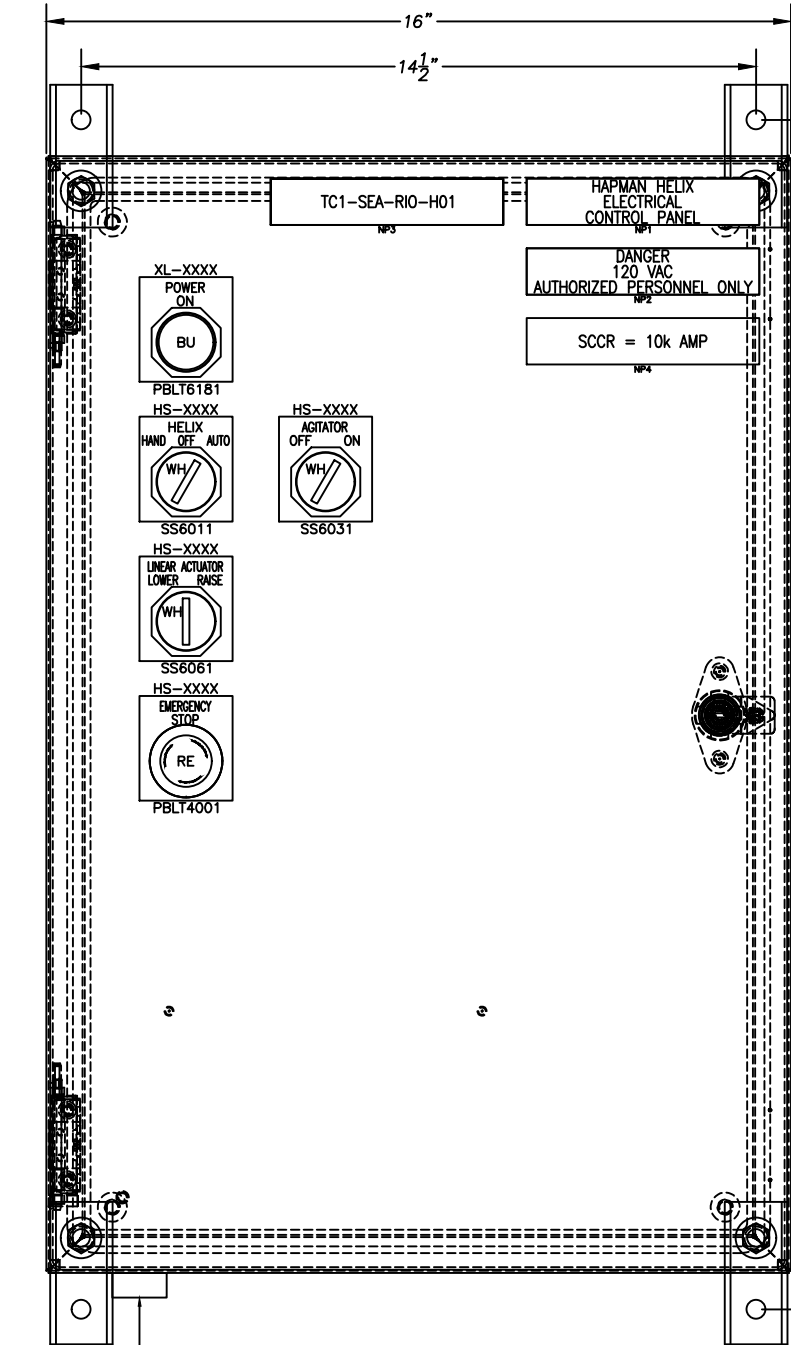
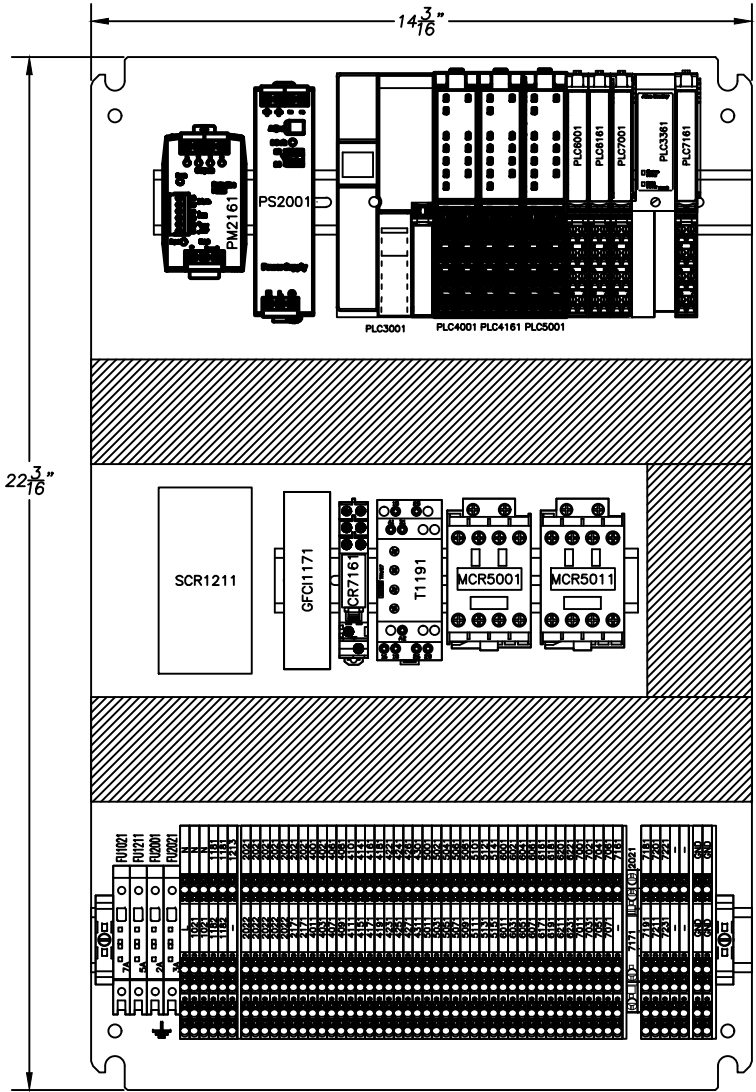
----- FIELD WIRING  
\_\_\_\_\_ PANEL WIRING



5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 349-2477

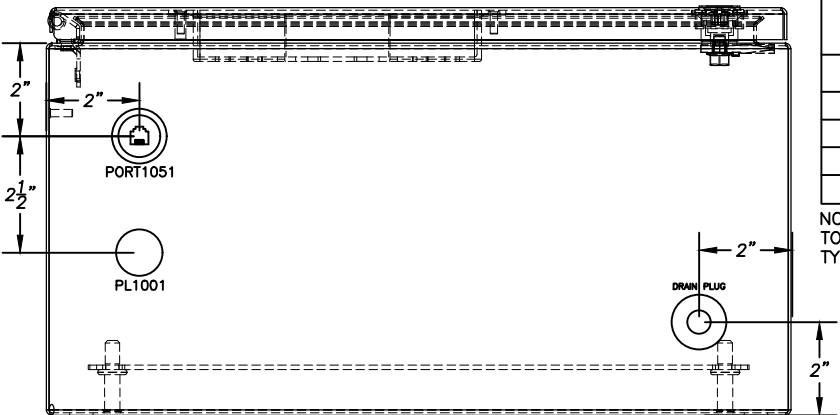
|                                            |         |         |           |          |              |          |
|--------------------------------------------|---------|---------|-----------|----------|--------------|----------|
| DATE                                       | 4/21/23 | JOB NO. | X20335-BC |          | DRAWN        | MRL      |
| SCALE                                      | N/A     |         | BOM = 1   | DWG = 1  |              |          |
| TITLE                                      |         |         |           |          |              |          |
| RESERVED<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |          |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |         |           |          |              |          |
| SHEET                                      | 12      | OF      | 17        | DWG. NO. | X20335BC-B00 | REV. 0.0 |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



| NAMEPLATE DESCRIPTIONS |                 |            |                           |
|------------------------|-----------------|------------|---------------------------|
| N.P. #                 | LINE 1          | LINE 2     | LINE 3                    |
| NP1                    | HAPMAN HELIX    | ELECTRICAL | CONTROL PANEL             |
| NP2                    | DANGER          | 120 VAC    | AUTHORIZED PERSONNEL ONLY |
| NP3                    | TC1-SEA-RIO-H01 |            |                           |
| NP4                    | SCCR = 10k AMPS |            |                           |

- NOTES:
- NP1 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - NP2 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" RED FRONT WITH 1/4" HIGH ENGRAVED WHITE LETTERS.
  - NP3-NP4 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - ALL OTHER NAMEPLATES SHALL BE 3/4"x2" WHITE FRONT WITH BLACK LETTERS



PIN/PLUG CONNECTIONS BY OTHERS

| FUSE REPLACEMENT |           |          |               |
|------------------|-----------|----------|---------------|
| FUSE #           | TYPE/SIZE | AMPERAGE | VOLTAGE       |
| FU1021           | MDL-7-R   | 7A       | 250VAC        |
| FU1211           | MDL-5-R   | 5A       | 250VAC        |
| FU2001           | MDL-2-R   | 2A       | 250VAC        |
| FU2021           | ABC-3-R   | 3A       | 250VAC/125VDC |

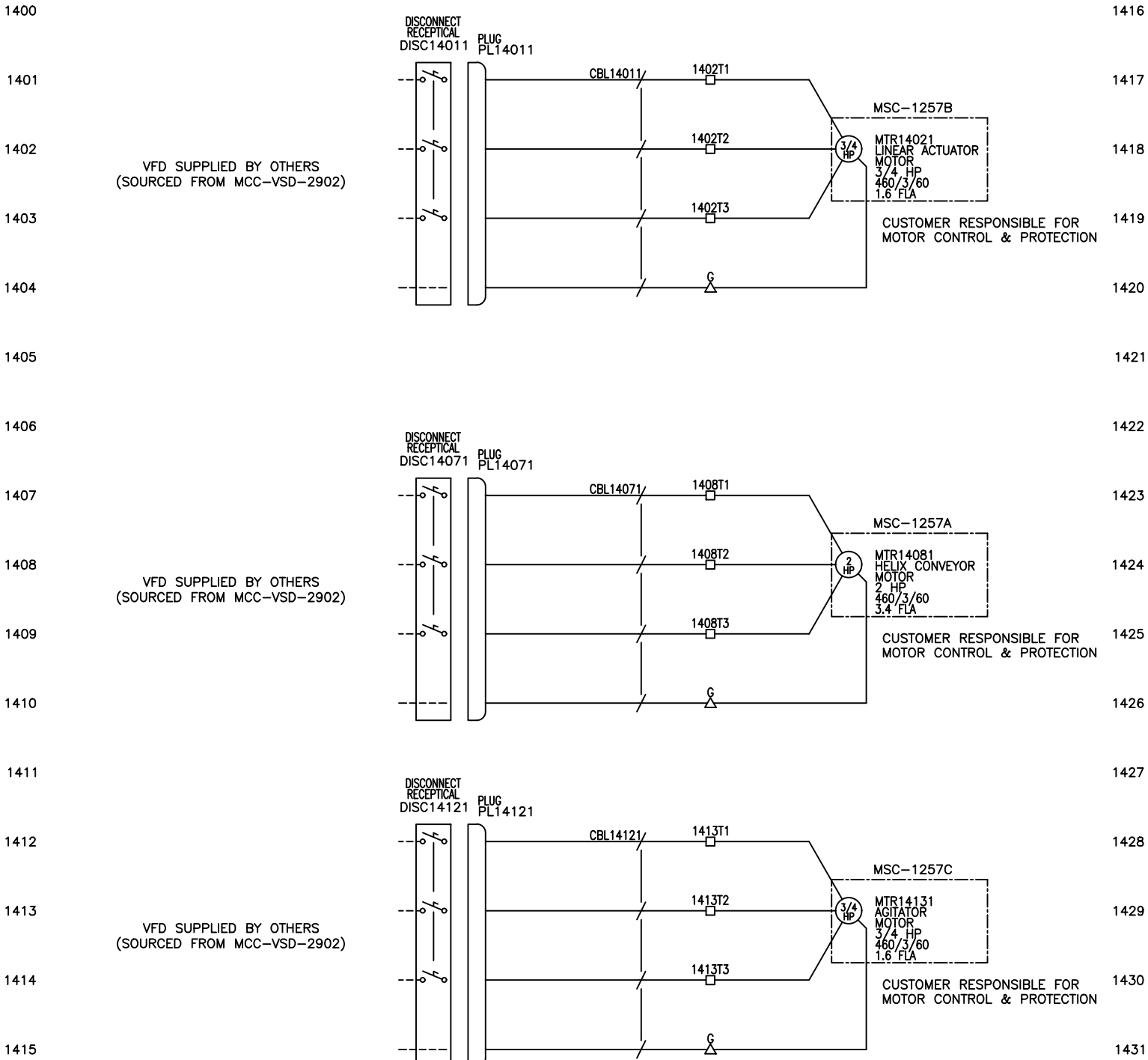
NOTE:  
TO REDUCE THE RISK OF FIRE - REPLACE ONLY WITH THE SAME TYPE AND RATING OF FUSE IN THE FUSE REPLACEMENT CHART.

|                    |                                  |
|--------------------|----------------------------------|
| VOLTAGE:           | 120 VOLT                         |
| PHASE & FREQ.:     | 1PH., 60HZ                       |
| FULL LOAD CURRENT: | 2 AMP                            |
| LARGEST MOTOR FLA: | 1 AMP                            |
| S.C.C.R.:          | 10k AMP                          |
| ENCLOSURE RATING:  | TYPE 4X                          |
| SCHEMATIC:         | X20335BC-B00 (MAIN PANEL)        |
| MFG. BY:           | KMC Global Controls & Automation |
| UL FILE NUMBER:    | E500032                          |

|                                            |                                                                     |                                                                                       |                |
|--------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------|
| <b>HAPMAN</b><br>Ideas that move           |                                                                     | 5944 E. N. AVE., KALAMAZOO, MICHIGAN 49048<br>PHONE (269) 343-1675 FAX (269) 349-2477 |                |
| DATE<br>4/21/23                            | JOB NO.<br>X20335-BC                                                | DRAWN<br>BOM = 1                                                                      | MRL<br>DWG = 1 |
| SCALE<br>N/A                               | TITLE<br><b>MAIN PANEL LAYOUT</b><br>TYPE 4X - HELIX CONVEYOR HI/LO |                                                                                       |                |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) | SHEET 13 OF 17                                                      |                                                                                       | REV. 0.0       |
|                                            | DWG. NO. X20335BC-B00                                               |                                                                                       |                |



| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



CONDUCTOR WIRE CODE

BLACK: GREATER THAN 120VAC  
RED: 120VAC OR LESS  
WHITE: GROUNDED AC LINE  
BLUE: UN GROUNDED VDC  
WH/BLU: GROUNDED VDC  
GREEN: GROUNDING CONDUCTORS  
ORANGE: CIRCUITS THAT REMAIN ENERGIZED  
WHEN MAIN DISC. IS "OFF"

WIRE GAUGE

MOTOR CIRCUITS: MIN. 12 AWG  
CONTROL CIRCUITS: MIN. 16 AWG  
PLC I/O: MIN. 18 AWG  
FUSES: LINE/LOAD SIDE  
SHALL BE EQUAL  
AWG  
RETURN/SUPPLY: SHALL BE EQUAL  
AWG

LEGEND

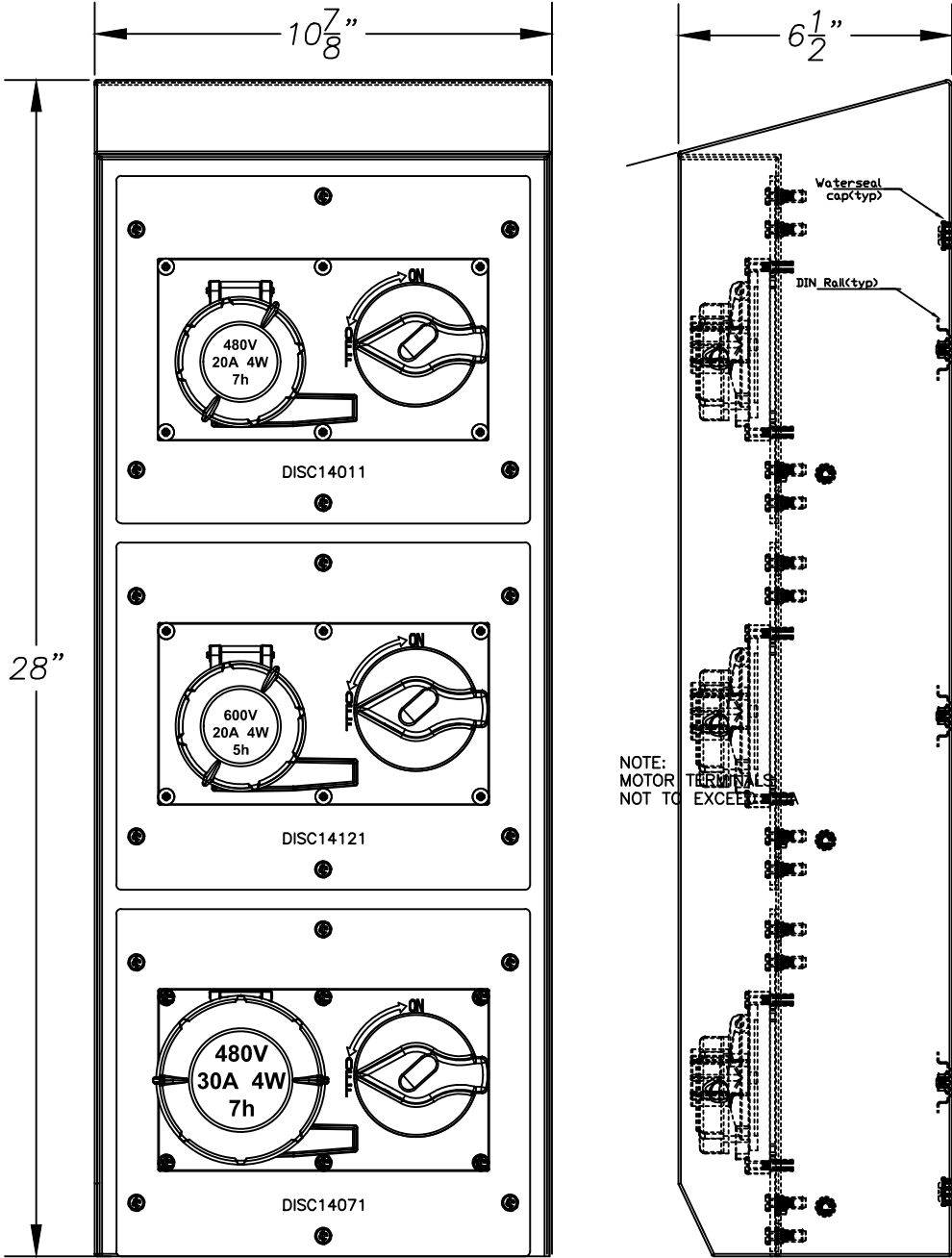
----- FIELD WIRING  
————— PANEL WIRING

**HAPMAN**  
Ideas that move

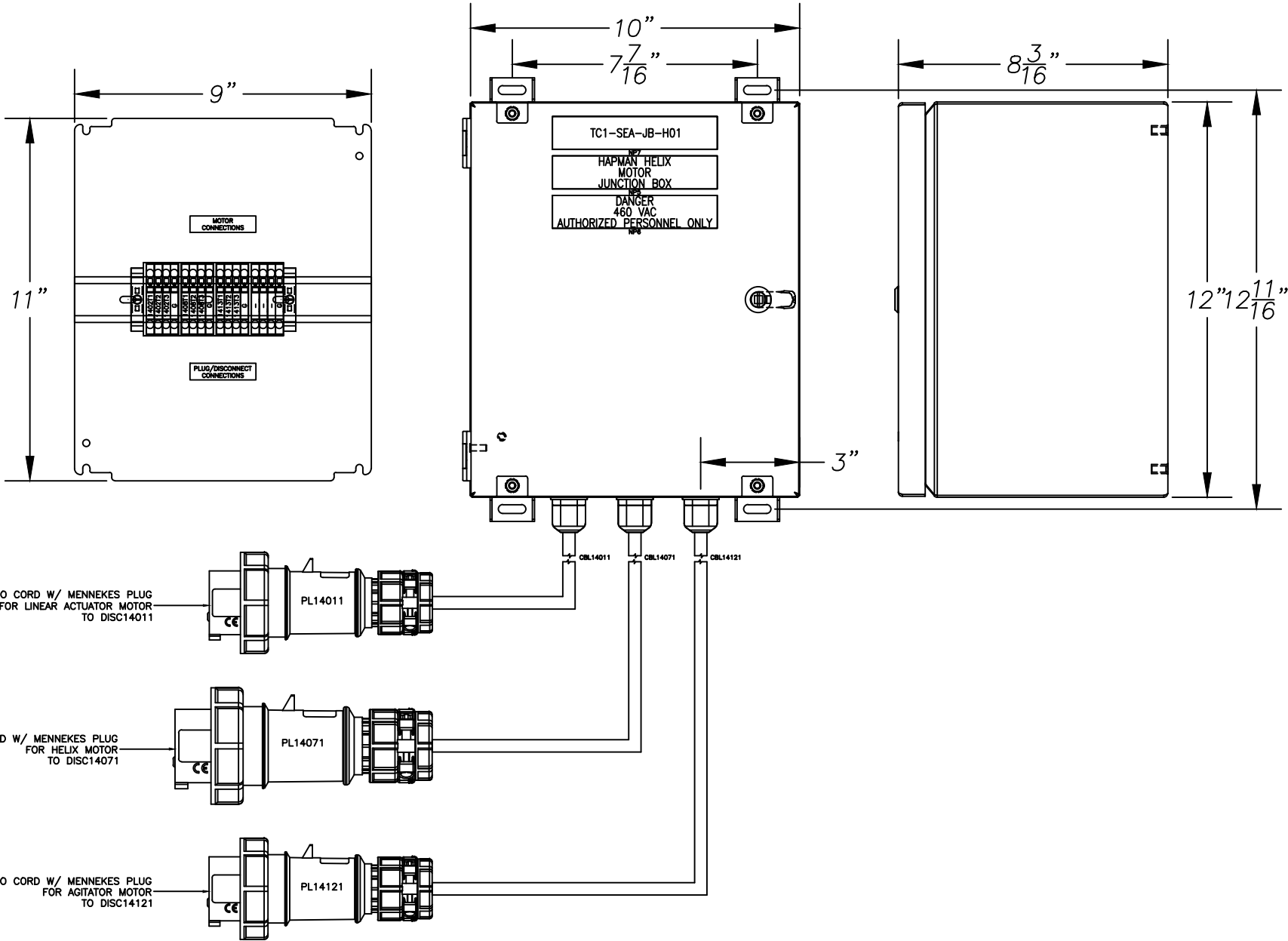
5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 349-2477

|                                                                     |         |         |           |          |              |
|---------------------------------------------------------------------|---------|---------|-----------|----------|--------------|
| DATE                                                                | 4/21/23 | JOB NO. | X20335-BC | DRAWN    | MRL          |
| SCALE                                                               | N/A     |         |           | BOM = 1  | DWG = 1      |
| TITLE                                                               |         |         |           |          |              |
| MOTOR JUNCTION BOX WIRING DIAGRAM<br>TYPE 4X - HELIX CONVEYOR HI/LO |         |         |           |          |              |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                          |         |         |           |          |              |
| SHEET                                                               | 14      | OF      | 17        | DWG. NO. | X20335BC-B00 |
|                                                                     |         |         |           |          | REV. 0.0     |

| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



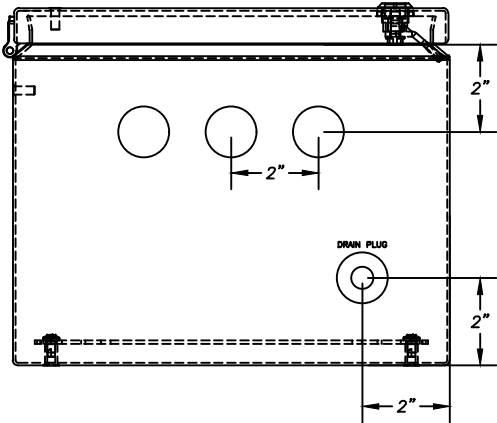
NOTE:  
MOTOR TERMINALS  
NOT TO EXCEED



10' 12/4 SO CORD W/ MENNEKES PLUG  
FOR LINEAR ACTUATOR MOTOR  
TO DISC14011

10' 12/4 SO CORD W/ MENNEKES PLUG  
FOR HELIX MOTOR  
TO DISC14071

10' 12/4 SO CORD W/ MENNEKES PLUG  
FOR AGITATOR MOTOR  
TO DISC14121



| NAMEPLATE DESCRIPTIONS |                |         |                           |
|------------------------|----------------|---------|---------------------------|
| N.P. #                 | LINE 1         | LINE 2  | LINE 3                    |
| NP5                    | HAPMAN HELIX   | MOTOR   | JUNCTION BOX              |
| NP6                    | DANGER         | 460 VAC | AUTHORIZED PERSONNEL ONLY |
| NP7                    | TC1-SEA-JB-H01 |         |                           |

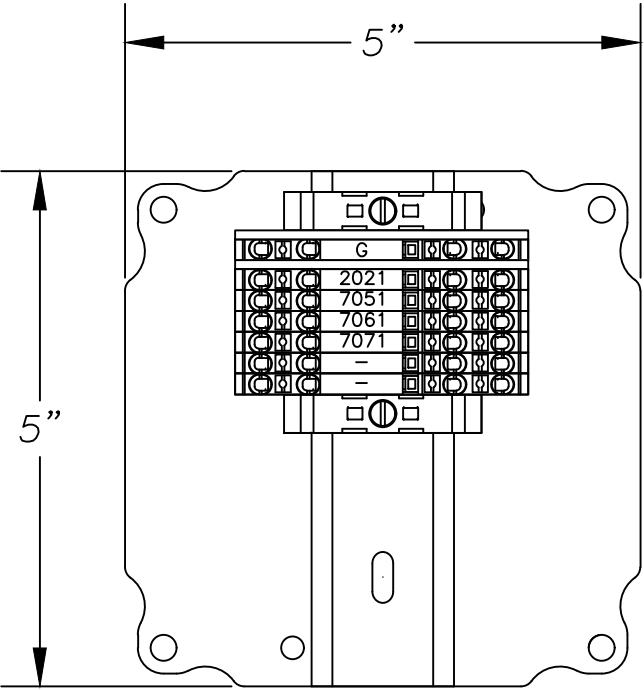
- NOTES:
- NP5 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - NP6 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" RED FRONT WITH 1/4" HIGH ENGRAVED WHITE LETTERS.
  - NP7 SHALL BE 1"x5" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.
  - ALL OTHER NAMEPLATES SHALL BE 3/4"x2" WHITE FRONT WITH BLACK LETTERS

| MOTOR JUNCTION BOX |                                  |
|--------------------|----------------------------------|
| VOLTAGE:           | 460 VOLT                         |
| PHASE & FREQ.:     | 3PH., 60HZ                       |
| ENCLOSURE RATING:  | TYPE 4X                          |
| SCHEMATIC:         | X20335BC-B00                     |
| MFG. BY:           | KMC Global Controls & Automation |
| UL FILE NUMBER     | E500032                          |

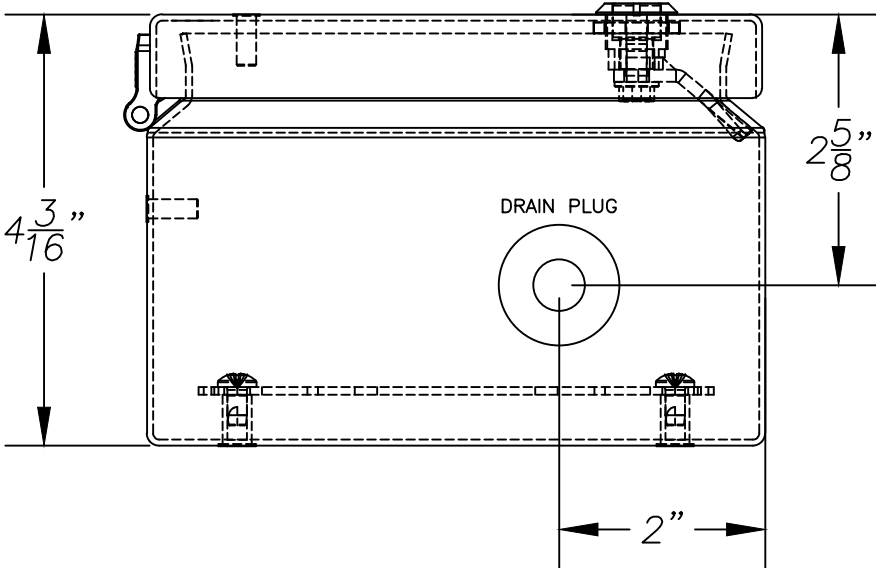
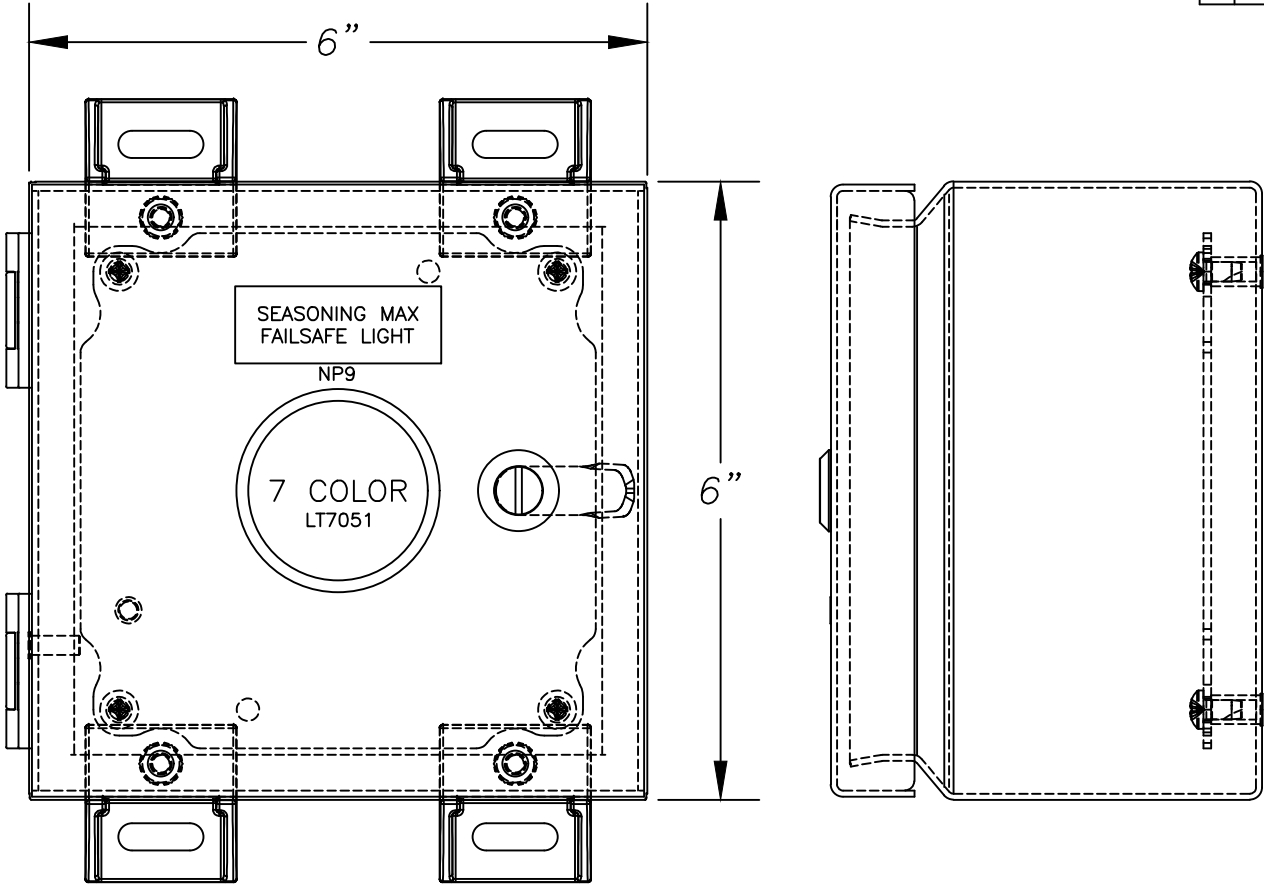
| HAPMAN                                     |         |              |           |
|--------------------------------------------|---------|--------------|-----------|
| ideas that move                            |         |              |           |
| 5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  |         |              |           |
| PHONE (269) 343-1675 FAX (269) 349-2477    |         |              |           |
| DATE                                       | 4/21/23 | JOB NO.      | X20335-BC |
| SCALE                                      | N/A     | DRAWN        | MRL       |
| SHEET                                      |         | BOM = 1      | DWG = 1   |
| TITLE                                      |         |              |           |
| MOTOR JUNCTION BOX                         |         |              |           |
| TYPE 4X - HELIX CONVEYOR HI/LO             |         |              |           |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO) |         |              |           |
| SHEET                                      |         | DWG. NO.     | REV.      |
| 15 OF 17                                   |         | X20335BC-B00 | 0.0       |



| REV | DESCRIPTION | DATE    | BY  |
|-----|-------------|---------|-----|
| 0.0 | AS BUILT    | 6/28/23 | MRL |



NOTE:  
USE COPPER WIRE ONLY WITH INSULATION  
RATING OF 75°C OR GREATER



| NAMEPLATE DESCRIPTIONS |               |                |        |
|------------------------|---------------|----------------|--------|
| N.P. #                 | LINE 1        | LINE 2         | LINE 3 |
| NP9                    | SEASONING MAX | FAILSAFE LIGHT | -      |

NOTES:  
1. NP9 SHALL BE 2"x3/4" LAMINATED 1/16" MICARTA "GRAVIPLY" WHITE FRONT WITH 1/4" HIGH ENGRAVED BLACK LETTERS.

| MULTICOLORED LIGHT JUNCTION BOX |                                  |
|---------------------------------|----------------------------------|
| VOLTAGE:                        | 24VDC                            |
| ENCLOSURE RATING:               | TYPE 4X                          |
| SCHEMATIC:                      | X20335BC-B00                     |
| MFG. BY:                        | KMC Global Controls & Automation |
| UL FILE NUMBER                  | E500032                          |

**HAPMAN**  
Idea that move

5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 348-2477

|                                                                                                |                       |          |         |
|------------------------------------------------------------------------------------------------|-----------------------|----------|---------|
| DATE 4/21/23                                                                                   | JOB NO. X20335-BC     | DRAWN    | MRL     |
| SCALE N/A                                                                                      |                       | BOM = 1  | DWG = 1 |
| TITLE<br><b>MULTICOLORED LIGHT JUNCTION BOX PANEL LAYOUT</b><br>TYPE 4X - HELIX CONVEYOR HI/LO |                       |          |         |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                                                     |                       |          |         |
| SHEET 16 OF 17                                                                                 | DWG. NO. X20335BC-B00 | REV. 0.0 |         |

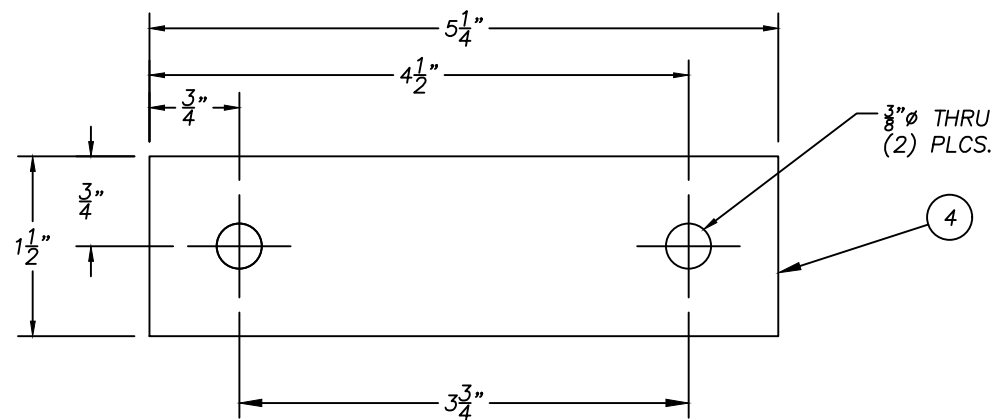
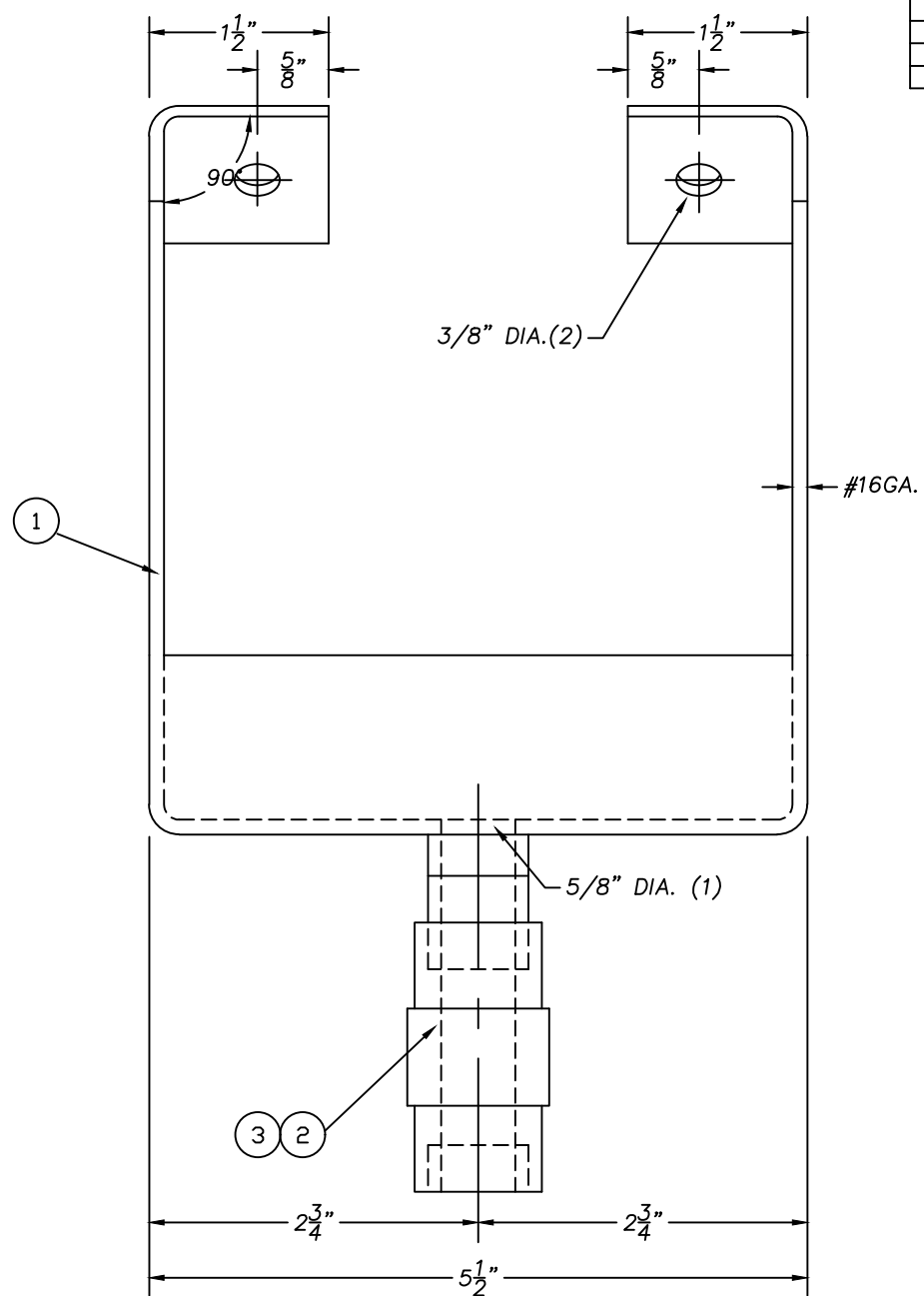
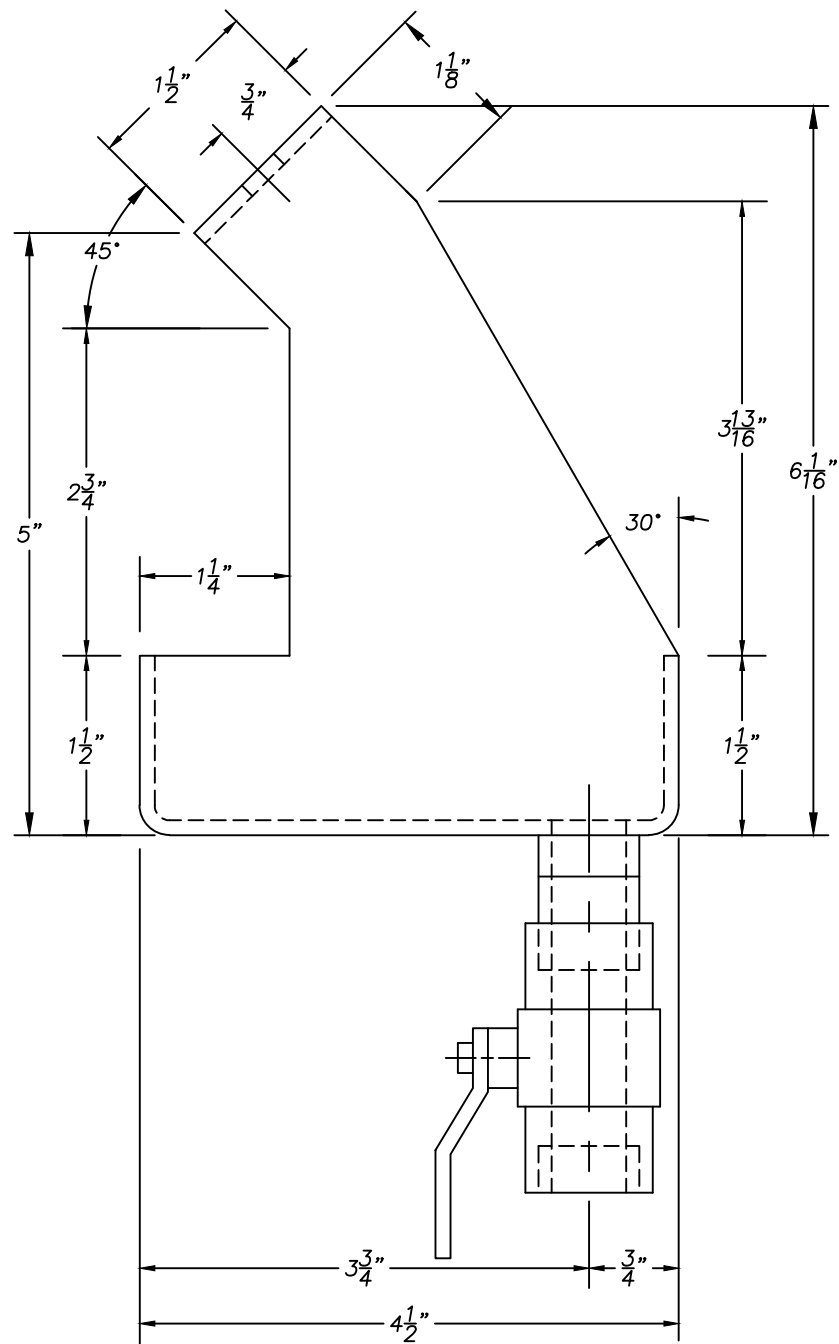
| ITEM | QTY | PART #         | MANUFACTURER | DESCRIPTION                                                            | DESCRIPTOR                |
|------|-----|----------------|--------------|------------------------------------------------------------------------|---------------------------|
| 1    | 1   | CSD24168SS     | HOFFMAN      | 24"x16"x8" TYPE 4X STAINLESS STEEL ENCLOSURE                           | MAIN PANEL                |
| 2    | 1   | CP2416         | HOFFMAN      | SUB PANEL                                                              | .                         |
| 3    | 1   | CMFKSS         | HOFFMAN      | TYPE 4X STAINLESS STEEL FOOT KIT                                       | .                         |
| 4    | 1   | IAEBHC6        | PANDUIT      | PROG. PORT, RJ45 CAT6                                                  | PORT1051                  |
| 5    | 1   | 576-S10-004    | QUIKTRON     | 4' SHIELDED CAT6 ETHERNET CABLE                                        | ENET-3041                 |
| 6    | 4   | 1492-H6        | AB           | 300V 12A FUSIBLE TERMINAL BLOCK #30-#12AWG                             | FU1021,1211,2001,2021     |
| 7    | 1   | 1492-N37       | AB           | FUSIBLE TERMINAL END BARRIER                                           | FU1021-2001               |
| 8    | 1   | MDL-7-R        | BUSSMANN     | 7A 1/4"x1-1/4" TIME-DELAY GLASS FUSE                                   | FU1021                    |
| 9    | 1   | MDL-5-R        | BUSSMANN     | 5A 1/4"x1-1/4" TIME-DELAY GLASS FUSE                                   | FU1211                    |
| 10   | 1   | MDL-2-R        | BUSSMANN     | 2A 1/4"x1-1/4" TIME-DELAY GLASS FUSE                                   | FU2001                    |
| 11   | 1   | ABC-3-R        | BUSSMANN     | 3A 1/4"x1-1/4" FAST-ACTING CERAMIC FUSE                                | FU2021                    |
| 12   | 1   | 1492-MCGAT115  | AB           | 15A 1 POLE GFCI CIRCUIT BREAKER                                        | GFCI1171                  |
| 13   | 1   | 1492-AMCDIN1   | AB           | GFCI CIRCUIT BREAKER DIN CLIP                                          | GFCI1171                  |
| 14   | 1   | TRW27          | EATON        | 24-240 VAC/DC MULTI-FUNCTION DPDT TIMER                                | T1191                     |
| 15   | 1   | 179502-A       | FMC          | 115/60/1 SCR                                                           | SCR1211                   |
| 16   | 1   | 4SSL1/2DR      | APPLETON     | 2"x4"x2-1/8" HANDYBOX                                                  | SCR1211                   |
| 17   | 1   | 180A           | APPLETON     | 4-1/8"x2-3/8" HANDYBOX COVER                                           | SCR1211                   |
| 18   | 1   | PIM60.245      | PULS         | 24VDC 2.5A NEC CLASS2 COMPLAINT POWER SUPPLY                           | PS2001                    |
| 19   | 1   | PISA11.CLASS2  | PULS         | 24VDC 14.8A PROTECTION MODULE                                          | PM2161                    |
| 20   | 1   | 1734-AENT      | AB           | POINT I/O ETHERNET ADAPTER                                             | PLC3001                   |
| 21   | 1   | 1734-FPD       | AB           | POINT I/O FIELD POTENTIAL DISTRIBUTOR MODULE                           | PLC3361                   |
| 22   | 2   | 1734-IB8S      | AB           | 8 POINT SAFETY SINKING INPUT CARD                                      | PLC4001,4161              |
| 23   | 1   | 1734-OB8S      | AB           | 8 POINT SAFETY SOURCING OUTPUT CARD                                    | PLC5001                   |
| 24   | 2   | 1734-IB8       | AB           | POINT I/O 8PT DIGITAL INPUT CARD                                       | PLC6001,6161              |
| 25   | 2   | 1734-OB8       | AB           | POINT I/O 8PT DIGITAL OUTPUT CARD                                      | PLC7001,7161              |
| 26   | 10  | 1734-TBS       | AB           | POINT I/O SPRING TERMINAL BASE                                         | PLC4001-7161              |
| 27   | 1   | 800H-FRXTQH2RA | AB           | 30mm 12-130VAC/DC RED ILLUMINATED PUSH/PULL MUSH HEAD PB               | PBLT4001                  |
| 28   | 1   | 800T-XA4       | AB           | 30mm 2 N.C. CONTACT BLOCK                                              | PBLT4001                  |
| 29   | 5   | CT-SPECIAL     | CT           | 30mm LEGEND CUSTOM TEXT (SEE DRAWING)                                  | PBLT4001,6181,SS6011-6061 |
| 30   | 2   | 700-CF400EJ    | AB           | 24VDC CONTROL RELAY WITH 4 N.O. CONTACTS                               | MCR5001,5011              |
| 31   | 1   | 800H-JR2A      | AB           | 30mm 3 POSITION MAINTAINED SELECTOR SWITCH                             | SS6011                    |
| 32   | 1   | 800H-HR2D1     | AB           | 30mm 2 POSITION MAINTAINED SELECTOR SWITCH                             | SS6031                    |
| 33   | 1   | 800H-JR91A     | AB           | 3 POS. SPRING RETURN SELECTOR SWITCH W/ 1 N.O. & 1 N.C. CONTATCS       | SS6061                    |
| 34   | 1   | 800H-QRBH2B    | AB           | 30mm 12-130VAC/DC BLUE ILLUM. FLUSH HEAD PB W/ 1 N.O. & 1 N.C. CONTACT | PBLT6181                  |
| 35   | 1   | 700-HK32Z24    | AB           | 24VDC DPDT SLIM RELAY CUBE                                             | CR7001                    |
| 36   | 1   | 700-HN222      | AB           | DPDT SLIM RELAY SOCKET                                                 | CR7001                    |
| 37   | 1   | 1492-LD4DR     | AB           | DOUBLE STACK DIODE TERMINAL                                            | D7171                     |
| 38   | 1   | 1492-EBLD4     | AB           | END PLATE FOR DOUBLE STACK TERMINAL                                    | D7171                     |
| 39   | 51  | 3210609        | PHOENIX      | 5.2mm DOUBLE STACK PUSH IN TERMINAL                                    | .                         |
| 40   | 2   | 3210611        | PHOENIX      | 5.2mm DOUBLE STACK PUSH IN GROUND TERMINAL                             | .                         |
| 41   | 4   | 3210613        | PHOENIX      | END PLATE FOR 5.2mm DOUBLE STACK PUSH IN TERMINAL                      | .                         |
| 42   | 2   | 0800886        | PHOENIX      | SCREW ON END STOP                                                      | .                         |
| 43   | 2'  | 111023         | E-RAIL       | 35mm DIN RAIL                                                          | .                         |
| 44   | 4'  | F2X4LG6        | PANDUIT      | 2"x4" WIREWAY                                                          | .                         |
| 45   | 4'  | C2LG6          | PANDUIT      | 2" WIREWAY COVER                                                       | .                         |
| 46   | 4   | JPPT4A         | CT           | 1"x5" CUSTOM ENGRAVED LEGEND                                           | NP1-4                     |
| 47   | 1   | SSTC 1CD       | CROUSE-HINDS | DRAIN PULG, 1/2" STAINLESS STEEL                                       | .                         |
| 48   | 1   | 212PA          | MENCOM       | 1/2" NPT LOCKNUT                                                       | .                         |
| 49   | 1   | PPS0305W2100   | PANDUIT      | ARC FLASH LABEL                                                        | .                         |
| 50   |     |                |              |                                                                        |                           |
| 51   |     |                |              |                                                                        |                           |
| 52   |     |                |              |                                                                        |                           |
| 53   |     |                |              |                                                                        |                           |
| 54   |     |                |              |                                                                        |                           |
| 55   |     |                |              |                                                                        |                           |
| 56   |     |                |              |                                                                        |                           |

| ITEM | QTY  | PART #         | MANUFACTURER  | DESCRIPTION                                                | DESCRIPTOR                      |
|------|------|----------------|---------------|------------------------------------------------------------|---------------------------------|
| 57   | 1    | 888N-M4AF1-1F  | AB            | 4 PIN 600V, 10A, 16 AWG MALE RECEPTACLE 1' LEAD            | PL1001                          |
| 58   | 1    | 889N-F4AFC-12F | AB            | 4 PIN 600V, 10A, FEMALE QUICK CHANGE CONNECTOR W/ 12' LEAD | RECEP1001 (SHIP LOOSE)          |
| 59   | REF  | 855T-B24TL4    | ALLEN BRADLEY | 24VDC STEADY LED TOWER LIGHT- (AB), RED                    | LT7011                          |
| 60   | REF  | 855T-BCBC      | ALLEN BRADLEY | SURFACE MOUNT BASE- (AB), W/CAP, 1/2 NPT                   | LT7011                          |
| 61   |      |                |               |                                                            |                                 |
| 62   |      |                |               |                                                            |                                 |
| 63   | 1    | 12108ELJSS     | SCE           | 12"x10"x8" TYPE 4X STAINLESS STEEL ENCLOSURE               | MOTOR JUNCTION BOX              |
| 64   | 1    | 12P10          | SCE           | SUB PANEL                                                  | .                               |
| 65   | 12   | 3211757        | PHOENIX       | 6.2mm 30A PUSH IN TERMINAL                                 | .                               |
| 66   | 4    | 3211766        | PHOENIX       | 6.2mm PUSH IN GROUND TERMINAL                              | .                               |
| 67   | 4    | 3030420        | PHOENIX       | END COVER FOR 6.2mm PUSH IN TERMINAL                       | .                               |
| 68   | 2    | 0800886        | PHOENIX       | SCREW ON END STOP                                          | .                               |
| 69   | 1'   | 111023         | E-RAIL        | 35mm DIN RAIL                                              | .                               |
| 70   | 3    | SHC1036        | HUBBELL       | 3/4" NPT MACHINED ALUMINUM CORD GRIP (0.5-0.63")           | CBL14011,14071,14121            |
| 71   | 3    | 20509002       | HUBBELL       | 3/4" NPT METAL CLAD SEALING O-RING                         | CBL14011,14071,14121            |
| 72   | 3    | 234PA          | MENCOM        | 3/4" NPT LOCKNUT                                           | CBL14011,14071,14121            |
| 73   | 33'  | 12/4 SO CORD   | CAROL         | 12/4 SO CORD                                               | CBL14011,14071,14121            |
| 74   | 1    | ME420P7W       | MENNEKES      | 20A 3P 4W 480V WATERTIGHT PLUG                             | PL14011                         |
| 75   | 1    | ME430P7W       | MENNEKES      | 30A 3P 4W 480V WATERTIGHT PLUG                             | PL14071                         |
| 76   | 1    | ME420P5W       | MENNEKES      | 20A 3P 4W 600V WATERTIGHT PLUG                             | PL14121                         |
| 77   | 1    | SPX3A-639068   | MENNEKES      | AMAXX STAINLESS STEEL 3 POSITION INTERLOCKED RECEPTACLE    | DISC14011,14071,14121           |
| 78   | 3    | JPPT4A         | CT            | 1"x5" CUSTOM ENGRAVED LEGEND                               | NP5-7                           |
| 79   | 1    | SSTC 1CD       | CROUSE-HINDS  | DRAIN PULG, 1/2" STAINLESS STEEL                           | .                               |
| 80   | 1    | 212PA          | MENCOM        | 1/2" NPT LOCKNUT                                           | .                               |
| 81   | 1    | PPS0305W2100   | PANDUIT       | ARC FLASH LABEL                                            | .                               |
| 82   |      |                |               |                                                            |                                 |
| 83   |      |                |               |                                                            |                                 |
| 84   |      |                |               |                                                            |                                 |
| 85   |      |                |               |                                                            |                                 |
| 86   |      |                |               |                                                            |                                 |
| 87   | 1    | 606ELJSS       | SCE           | 6"x6"x4" TYPE 4X STAINLESS STEEL ENCLOSURE                 | MULTICOLORED LIGHT JUNCTION BOX |
| 88   | 1    | 6P6JGALV       | SCE           | GALVANIZED SUB PANEL                                       | .                               |
| 89   | 1    | K50L2FRGB7QP   | BANNER        | 50mm 10-30VDC 7 COLOR RGB PILOT LIGHT                      | LT7051                          |
| 90   | 1    | MQDC1-515RA    | BANNER        | CORDSET, 5 PIN STRAIGHT CONNECTOR 15'                      | LT7051                          |
| 91   | 1    | JPPT2A         | CT            | 3/4"x2" CUSTOM ENGRAVED LEGEND (WHITE W/ BLACK)            | NP8                             |
| 92   | 6    | 1492-P3Q       | AB            | 5.1mm 4-WIRE PUSH IN TERMINAL                              | .                               |
| 93   | 1    | 1492-PG3Q      | AB            | 5.1mm 4-WIRE PUSH IN GROUND TERMINAL                       | .                               |
| 94   | 2    | 1492-EBP3Q     | AB            | END PLATE FOR 5.1mm 4 WIRE PUSH IN TERMINAL                | .                               |
| 95   | 2    | 0800886        | PHOENIX       | SCREW ON END STOP                                          | .                               |
| 96   | 0.5' | 111023         | E-RAIL        | 35mm DIN RAIL                                              | .                               |
| 97   | 1    | SSTC 1CD       | CROUSE-HINDS  | DRAIN PULG, 1/2" STAINLESS STEEL                           | .                               |
| 98   | 1    | 212PA          | MENCOM        | 1/2" NPT LOCKNUT                                           | .                               |
| 99   | 1    | 31622003L      | HUBBELL       | 1/2" NPT LOCKING NUT                                       | .                               |
| 100  |      |                |               |                                                            |                                 |
| 101  |      |                |               |                                                            |                                 |
| 102  |      |                |               |                                                            |                                 |
| 103  |      |                |               |                                                            |                                 |
| 104  |      |                |               |                                                            |                                 |
| 105  |      |                |               |                                                            |                                 |

**HAPMAN**  
Ideas that move

5944 E. N AVE., KALAMAZOO, MICHIGAN 49048  
PHONE (269) 343-1675 FAX (269) 349-2477

|                                                                     |                          |                  |                |
|---------------------------------------------------------------------|--------------------------|------------------|----------------|
| DATE<br>4/21/23                                                     | JOB NO.<br>X20335-BC     | DRAWN<br>BOM = 1 | MRL<br>DWG = 1 |
| TITLE<br><b>BILL OF MATERIALS</b><br>TYPE 4X - HELIX CONVEYOR HI/LO |                          |                  |                |
| CUST. PROJECT SERVICES (FRITO LAY MODESTO)                          |                          |                  |                |
| SHEET<br>17 OF 17                                                   | DWG. NO.<br>X20335BC-B00 |                  | REV.<br>0.0    |



| REV. | DESCRIPTION | DATE | BY |
|------|-------------|------|----|
| 1    |             |      |    |
| 2    |             |      |    |
| 3    |             |      |    |
| 4    |             |      |    |
| 5    |             |      |    |

|                                                                                                             |             |                                                     |           |
|-------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------|-----------|
| 5                                                                                                           |             |                                                     |           |
| 4                                                                                                           |             | SPACER PLATE- 3/8in THK x 1 1/2 x 5 1/4in, SS       | 1         |
| 3                                                                                                           |             | BALL VALVE- SS VALVE, TEFLON SEAT, SS BALL, 1/2 NPT | 1         |
| 2                                                                                                           |             | CLOSE NIPPLE- 1/2 NPT, SS                           | 1         |
| 1                                                                                                           |             | DRIP PAN- 16GA, SS                                  | 1         |
| ITEM NO.                                                                                                    | DRAWING NO. | DESCRIPTION                                         | QUAN      |
| <b>HAPMAN</b><br>Ideas that move                                                                            |             |                                                     |           |
| P.O. BOX 2321<br>6002 E. KILGORE ROAD, KALAMAZOO, MICHIGAN 49048<br>PHONE (269) 343-1675 FAX (269) 349-2477 |             |                                                     |           |
| DATE                                                                                                        | 6/11/13     | JOB NO.                                             |           |
| SCALE                                                                                                       | FULL        | DRAWN                                               | M. PORTER |
|                                                                                                             |             | BOM                                                 | 0         |
|                                                                                                             |             | DWG                                                 | 1         |
| TITLE                                                                                                       |             |                                                     |           |
| DRIVE DRIP PAN<br>#16 GA. STAINLESS STEEL                                                                   |             |                                                     |           |
| CUST.                                                                                                       |             | C:\HAPMAN\LIBRARY                                   |           |
| SHEET                                                                                                       | 1 OF 1      | DWG. NO.                                            | 9662-C    |
|                                                                                                             |             | REV.                                                |           |

| UNLESS OTHERWISE SPECIFIED<br>DIMENSIONS ARE ENGLISH (FEET AND/OR INCHES)<br>TOLERANCES |             |                               |        |
|-----------------------------------------------------------------------------------------|-------------|-------------------------------|--------|
| DECIMALS                                                                                | INCHES      | FEET                          | ANGLES |
| .X ± .030                                                                               |             |                               |        |
| .XX ± .015                                                                              |             |                               |        |
| .XXX ± .005                                                                             |             |                               |        |
|                                                                                         | X" = ±1/16" | X" - X" = ±1/4" UP TO 20 FEET | ±1°    |
|                                                                                         |             | X" - X" = ±1/2" OVER 20 FEET  |        |
| DO NOT SCALE DWG.<br>BREAK ALL SHARP EDGES                                              |             |                               |        |
| MATERIAL                                                                                |             |                               |        |
| FINISH                                                                                  |             |                               |        |